1991ApJ...379..482K

THE ASTROPHYSICAL JOURNAL, 379:482–506, 1991 October 1
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POWER-SPECTRUM ANALYSIS OF ONE-DIMENSIONAL REDSHIFT SURVEYS

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Received 1990 September 27; accepted 1991 April 18

ABSTRACT

We consider the statistical properties of power spectra obtained from "pencil-beam" redshift surveys, with particular attention to the recent study by Broadhurst et al. (BEKS). We discuss tests of the null hypothesis that the long-wavelength one-dimensional power spectrum found in such surveys is generated by the known small-scale clustering of galaxies.

The visual impression of the BEKS redshift data is dominated by a relatively small number of rich clumps, and we first test whether there is any evidence for nonuniformity in the distribution of these objects. We make a "clump catalog," identifying the richest clumps by eye from the redshift histogram of BEKS. The real clumps appear to have a width in redshift of about 800 km s^{-1}; if clumps in synthetic catalogs are given a similar width, this causes the power spectrum to fall quite rapidly with increasing frequency. This can enhance the apparent significance of any spike at $k \lesssim 0.1h$ Mpc^{-1} compared with the exponential power distribution seen at higher k. We generate an ensemble of mock catalogs with a Poissonian distribution of clumps, using the observed clump richness distribution, and find that power spectra with features as striking as the real data arise quite frequently.

To a good approximation, the Fourier transform of the one-dimensional galaxy counts is a complex Gaussian random field, and the statistical properties of the low-k power spectrum are calculable in terms of the survey selection functions and the power spectrum of small-scale galaxy clustering. This provides a prediction for the one-dimensional power level which agrees quite well with the Poisson clump model prediction, and allows us to calculate the frequency of peaks in the power spectrum, as well as such quantities as the uncertainty in the power estimated from a finite bandwidth. We generate realizations of power spectra and compare with the surveys used by BEKS. Somewhat surprisingly, we find that the power spectra for the two deep surveys alone show little evidence for periodicity or extra large-scale power—the visual impression of the counts notwithstanding. With the inclusion of the shallower samples the significance of the spike at $k \simeq 0.05h$ Mpc^{-1} increases but is still marginal (such peaks are expected to occur $\simeq 6\%$ of the time). The texture of our simulations matches very well with observation and shows that (often quite isolated) spikes are common features. We estimate the level of extra large-scale power which might be detectable with one-dimensional surveys of this kind, and find that it is rather large, although a factor of ~ 2 improvement in sensitivity can be obtained by a nonlinear clipping of the data. The limited sensitivity of these surveys means that the results are expected to be consistent with a wide range of models for large-scale structure. Most importantly, they are certainly consistent with a galaxy distribution which is very nearly uniform on 100 Mpc scales.

Subject headings: cosmology — galaxies: clustering — galaxies: redshift

1. INTRODUCTION

In a recent paper, Broadhurst et al. (1990, hereafter BEKS) have presented a compilation of some deep "pencil-beam" redshift surveys. Their redshift distribution displays a striking apparent periodicity on a scale of $128h^{-1}$ Mpc, and the power spectrum of the counts shows a sharp spike at the corresponding wavenumber. There are two important questions raised by these results. First, do the data provide evidence for substantial power on large scales? Second, does the apparent periodicity in the sample reflect some new phenomenon which lies outside our current ideas about the distribution of galaxies? These two questions are somewhat independent. If the answer to the first were positive, this would be important but, by itself, perhaps not too earth-shattering. If, however, the spike in the power spectrum is incompatible with the usual Copernican assumption of statistical isotropy and homogeneity of galaxy clustering, the consequences would be much more radical.

A casual inspection of the redshift data gives the dominant visual impression of strong clumps of galaxies, separated by nearly empty "voids." This is not surprising, since the beams are very narrow ($\sim 10'$ in radius, corresponding to about $2h^{-1}$ Mpc at the redshift of ~ 0.2, where dn/dz peaks), and galaxies are well known to be strongly clumped on these scales. BEKS have emphasized that the significance of features in the one-dimensional power spectrum depend crucially on the clumpiness of galaxies at small scales; the one-dimensional spectrum at some frequency k_1 is a projection of the three-dimensional power from frequencies greater than k_1, so even in the absence of any significant power on large scales, one still expects fluctuations in the very long wavelength one-dimensional power spectrum. BEKS estimated this contribution from small-scale clustering in two ways. The first method takes as input the known two-point function and the geometry of the survey, and allows one to predict the fluctuations in ξ or the long-wavelength one-dimensional power spectrum. The second method exploits the fact that at sufficiently long wavelengths the

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power spectrum should be exponentially distributed with a constant (white-noise) expected power. One can therefore ask whether the distribution of power seen (and, in particular, the large positive excursion in the spike) is consistent with such a distribution.

Both these methods rejected the null hypothesis quite convincingly ($p \lesssim 10^{-4}$). While we firmly endorse the ideas behind these tests, the importance of such a conclusion makes it seem worth looking into matters in more detail. In the first method, the exact level of low-k power depends on the details of the survey geometry, which BEKS took to be simply a uniform cylinder. Reality is more complicated, and we wanted to know how robust the answer is. In the second approach it is necessary to use a rather wide bandwidth to estimate the "scale height" of the power spectrum—one is effectively comparing the height of the spike at low k with the mean power determined largely at higher frequencies. The problem here is that, at high spatial frequencies, the power level can be suppressed by the finite width of the clumps in redshift space. If the power expected for realistic clump widths is appreciably falling over the bandwidth used, this will cause one to overestimate the significance of peaks at low k. Since any significance levels estimated in this way depend exponentially on the mean power, these points of concern merit detailed examination.

In this paper we will describe a number of calculations we have made in order to explore these effects. First, in § 2, we will use the redshift histogram from BEKS to construct a synthetic "clump catalog," with the aim of testing whether the clumps are Poisson-distributed. The power spectrum at low k is insensitive to the shape of the clumps, but if the clumps are given zero width in redshift, the predicted power at high k is much greater than the observed value. However, the observed clumps clearly do have an appreciable width; if the synthetic clumps are broadened appropriately, the power spectrum falls substantially even at quite low wavenumbers. We then generate an ensemble of clump catalogs with a Poisson distribution of clumps. These are subjected to power-spectrum analysis, and the simulated surveys and power spectra can be compared with the results of BEKS. In § 3 we have tried to elaborate on BEKS's method 1: i.e., we try to estimate what power level one would expect based on the known two-point function of galaxies, but using a more realistic model for the survey geometry and radial select function. Within the uncertainties of the calculation, this gives results which are consistent with those from the Poisson clump reshuffling.

Having calculated the expected power, we then look at the properties of *peaks* in power spectra, using a mixture of analytical and numerical techniques. Because this section may prove useful in other astronomical applications, we have presented the material in some detail. From these results, we are able to assign a revised significance level to the peak in the BEKS power spectrum (§ 4). This does not appear to be excessively high by comparison with the expectation of a small-scale clustering null hypothesis. We then ask: What level of large-scale power would be detectable in such a survey? We find that this is much larger than the "extra" large-scale power (relative to cold dark matter [CDM] predictions) found in recent three-dimensional surveys. In § 4.2 we show that some improvement in sensitivity to long-wavelength power can be gained by down-weighting the rich clumps. Finally, in anticipation of improved future surveys, we discuss how coherent peculiar velocities associated with galaxy clustering can, under certain circumstances, introduce interesting features in the one-dimensional power.

2. POISSON CLUMP MODEL

2.1. *Power Spectra and Random Walks*

In this section we calculate the power spectrum expected for a one-dimensional redshift survey under the most restrictive null hypothesis of no clustering whatsoever (Poisson-distributed points). To calculate the discrete approximation to the Fourier coefficients of $\delta n/n$ in a set of N galaxies, we evaluate

$$f_k = \sum_j N^{-1} \exp(ikx_j) \tag{2.1}$$

(with this definition, $|f_k|^2$ is the contribution to the fractional density variance from a single mode). In the absence of clustering, these coefficients execute a random walk on the Argand plane. The first and second moments of the power are evaluated by splitting one-dimensional space into a large number of cells with occupation numbers n_i (cf. Peebles 1980):

$$\begin{aligned} N^2\langle |f_k|^2\rangle &= \sum n_i^2 , \\ N^4\langle |f_k|^4\rangle &= 2[\sum n_i^2]^2 - \sum n_i^4 . \end{aligned} \tag{2.2}$$

For uncorrelated "shots," $n_i = 0$ or 1, and the moments become

$$\begin{aligned} \langle |f_k|^2\rangle &= 1/N , \\ \langle |f_k|^4\rangle &= (2N^2 - N)/N^4 . \end{aligned} \tag{2.3}$$

For $N \gg 1$ these become consistent with the Rayleigh distribution expected from the central-limit theorem:

$$P(|f_k|^2 > X) = \exp(-NX) . \tag{2.4}$$

For a selection function which is nonuniform, this distribution will not apply to the first few modes, where the transform of the selection function is important. At larger k, though, the selection function generally has little structure and Poisson power should be correct.

The assumption of a Poissonian galaxy distribution is clearly false, however. It is obvious to the eye that the BEKS galaxies are not randomly distributed; the visual impression of the data is dominated by the positions of relatively few rich clumps. If particles come in clumps, $n_i > 1$ is possible and the power can then be much greater. In evaluating the significance of a power-spectrum peak, it is therefore vital to assess correctly the true number of degrees of freedom in the data. To get an idea of the effect such clumping may have on the power spectrum, we have made a rough list of the most obvious clumps in the combined shallow and deep surveys. From Figures 1*a* and 1*b* of BEKS, we count 29 clumps, defined rather loosely as those which lie clearly above their selection

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TABLE 1

CLUMPS IN THE BEKS DATA

"Redshift"	n	"Richness"
−0.440..........	1	11.8
−0.345..........	3	12.1
−0.330..........	2	7.0
−0.275..........	7	15.6
−0.235..........	2	3.3
−0.208..........	11	15.4
−0.160..........	12	12.5
−0.115..........	9	7.4
−0.103..........	7	5.4
−0.075..........	9	6.2
−0.058..........	15	9.6
−0.040..........	13	7.8
−0.020..........	10	5.6
0.022..........	9	4.6
0.028..........	26	13.2
0.050..........	2	1.0
0.064..........	16	7.8
0.078..........	30	14.6
0.117..........	3	1.5
0.125..........	39	19.7
0.175..........	17	9.5
0.232..........	18	12.0
0.242..........	23	15.9
0.275..........	10	8.0
0.310..........	5	4.8
0.320..........	6	6.1
0.330..........	7	7.5
0.410..........	7	13.0
0.455..........	10	26.8

function curve; these are listed in Table 1 (the "richness" column will be explained shortly). Although a variety of different lists of this type could be produced, the results of this section depend mainly just on the size of the largest few clumps (since these dominate Σn_i^2), and are insensitive to the details of the catalog. We now ask what power is expected on a hypothesis where these clumps are Poisson-distributed. Our list accounts for a total of 329 galaxies, whereas BEKS analyzed a total of 396, of which 380 lay at $z < 0.5$ (T. J. Broadhurst 1991, private communication); we therefore should add in 51 single galaxies—i.e., clumps of size unity. The expected power becomes

$$\langle |f_k|^2 \rangle = \sum n_i^2/N^2 = 0.042 . \tag{2.5}$$

This is a factor of 16 larger than the number expected if all galaxies were independent. Although the number of degrees of freedom is thus much smaller than 380, the central-limit theorem should still be a good approximation: considering the above moments, we get $\langle |f_k|^4 \rangle / \langle |f_k|^2 \rangle^2 = 1.88$ as against the asymptotic 2. We will exploit this in § 3.

In our units, the height of the BEKS power spectrum peak at $128h^{-1}$ Mpc is 0.21, and is therefore roughly 5 times our mean estimated power level from small-scale clumpiness. BEKS estimated the peak to be 8 times the mean, so their estimate of the mean power would appear to be on the low side. The probability that a randomly chosen point on the power spectrum exceeds this height is $p = \exp(-0.21/0.042) = 0.007$, i.e., one would expect the power to exceed 5 times the mean about 1% of the time. Since the peak found is quite narrow, this seems to match quite well with the prediction of the null hypothesis. Although our clump catalog does not claim to be a precise representation of the BEKS data, this does illustrate that a power-spectrum peak of the observed amplitude need not be especially improbable.

2.2. *Clump Simulations*

We can illustrate this by making some simulations based on our clump list as a useful approximate representation of the BEKS data. We proceeded as follows:

1. Create an average redshift distribution (the "selection function") by smoothing the clump redshift distribution (using a Gaussian filter with $\sigma = 0.1$ in redshift). We work in comoving distance with $\Omega = 1$, so that a total size of the box $z = \pm 0.5$ is $2202h^{-1}$ Mpc.
2. Divide the observed number of galaxies per clump by the smoothed dp/dz to get a "richness" (third column of Table 1).
3. Select a number of clumps using a Poisson distribution with mean 29, and place these clumps randomly in the range $z = \pm 0.5$ (the number distribution of clumps appears close to uniform in redshift, so we take this for simplicity). For each clump, select randomly one of the observed richness values, and multiply richness by selection function to get the "observed" number of galaxies.
4. Select a number of "field" galaxies from a Poisson distribution of mean 51.
5. Scale the total clump + field numbers to 380 (the observed number being set by telescope time available), and generate random redshifts for the revised number of field galaxies.

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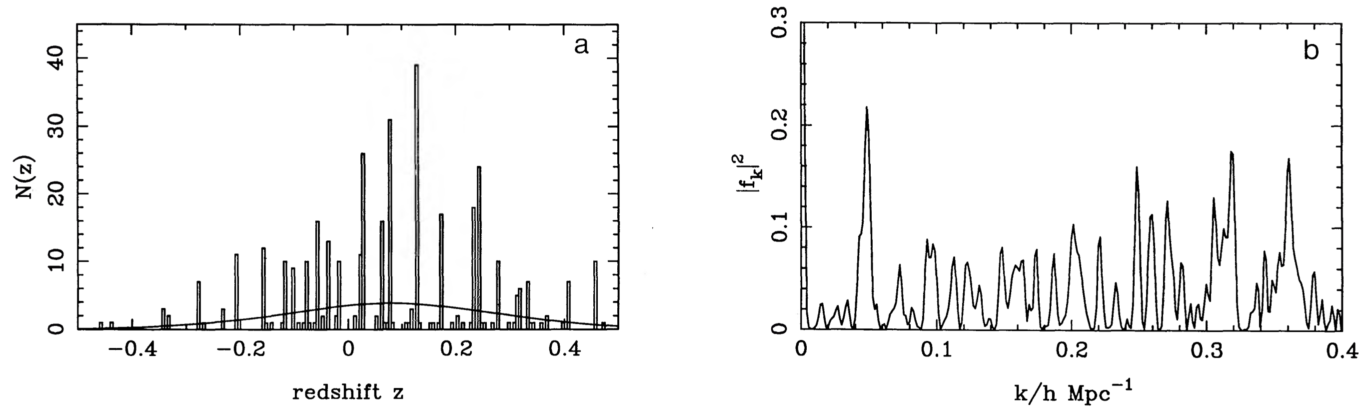

FIG. 1.—(*a*) Redshift distribution and (*b*) power spectrum of our clump list plus 51 random galaxies. The power-spectrum units are such that $|f_k|^2$ is the contribution to the variance in $\delta n/n$ from a given mode. Negative values of k have not been considered. Although the units differ, the power spectrum covers the same range of k as the BEKS plot. Note that here the amplitude of the power spectrum seems similar at low and high k, unlike the real data.

Before showing the results of any such simulation, let us look at the master data set. Figure 1 shows the redshift distribution (clumps + 51 random galaxies), together with its power spectrum. The power spectrum is a good likeness to that found by BEKS at low wavenumbers ($k \lesssim 0.1$, say), and there is a peak of about the right height at $k \simeq 0.05h$ Mpc^{-1}. However, the power at higher frequencies is much larger than that seen in reality. The reason for this is simple: the clumps in Figure 1 do not include any dispersion in the redshifts of the clump galaxies. The real clumps, in contrast, are clearly broadened. Adding a random dispersion (Gaussian-distributed with rms of 800 km s^{-1}) to the simulation produces Figure 2. This has a visual appearance which is more similar to the BEKS data in both redshift distribution and power spectrum. In particular, it is clear that the power spectrum is quite strongly suppressed below the asymptotic $k \to 0$ value for $k \gtrsim 0.1$: the highest peak in the power spectrum is most likely to appear at $\lambda \gtrsim 50h^{-1}$ Mpc. The power spectrum found by BEKS appears to be somewhat less attenuated at high k than our model, suggesting that in reality the situation is more complicated than this very simple model in which all clumps have the same size. However, this does not alter the qualitative point that peculiar velocities must reduce the power at high k.

Is the value of 800 km s^{-1} for the rms width of the clumps reasonable under our null hypothesis? The width of the clumps has three components: (1) width in real space; (2) velocity dispersion; and (3) redshift errors. The relative importance of these three factors is not easy to estimate; fortunately, however, all we need to know is the empirical total width. The similarity between Figure 2*a* and the binned counts in BEKS suggests that something close to our adopted figure is appropriate, but it would be nice to have an independent estimate. The one-dimensional power spectrum is the Fourier transform of the pair count as a function of redshift separation. Davis & Peebles (1983) give plots of $\xi(r_p, \pi)$—the two-point function as a function of redshift separation π—for various ranges of projected separation r_p, estimated from the CfA redshift survey. Each of these plots is well described by an exponential $\xi = a \exp(-\gamma\pi)$, though the coefficients a, γ depend on r_p. For the range $3.2h^{-1}$ Mpc $< \pi < 6.4h^{-1}$ Mpc, the "scale height" of the exponential is $1/\gamma \simeq 530$ km s^{-1}. The contribution to the power spectrum from the pairs with this range of projected separation should then fall off as $P(k) \propto 1/(k^2 + \gamma^2)$, and would be down by a factor of 2 at $k = \gamma \simeq 0.2h$ Mpc^{-1}. This is not a negligible effect, and would need to be taken into account, but it is certainly much smaller than the width we have assumed [where $P(k)$ is suppressed

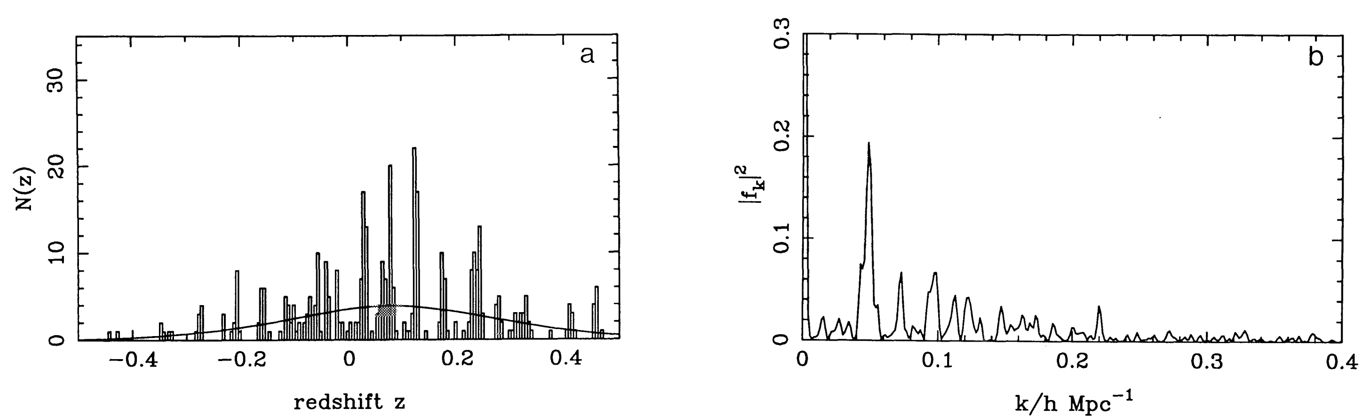

FIG. 2.—As in Fig. 1, except that a random velocity with dispersion 800 km s^{-1} has been added to the redshifts. Note that this greatly reduces the amplitude of the power spectrum for $k \gtrsim 0.1h$ Mpc^{-1}. The actual power appears to be less suppressed than our simple velocity smoothing would predict. This is understandable, since there is more substructure in the real clumps than in the smoothed simulation. The moral we would draw from Figs. 1 and 2 is that it is dangerous to use the power at $k \gtrsim 0.1h$ Mpc^{-1} to estimate the power from small-scale clustering. If one restricts attention to $k < 0.1h$ Mpc^{-1}, however, the mean power is uncertain because of the small number of independent estimates of P, and one must proceed either by modeling the clump richness distribution (§ 2) or by using independent estimates of small-scale galaxy clustering (§ 3).

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by a factor e at $k = 0.1$]. One possibility is that the broadening is due to redshift errors, and another is that the fields used—one of five in the south and one in three of the north—may contain abnormally rich and broad clumps. For the south Galactic pole (SGP) the estimated redshift error is too small to broaden the clumps by 800 km s^{-1}, but the redshifts are published (Broadhurst, Ellis, & Shanks 1988, hereafter BES), and so it is possible to get a clearer picture of the redshift pair distribution. For the SGP field, the distribution has a broad peak at the origin, and seems to be quite compatible with a clump width of 800 km s^{-1} [i.e., $\xi(\pi)$ appears to have a width $\simeq 2^{1/2} \times 800$ km s^{-1}). However, the four other fields in the BES survey which were not used by BEKS have a much narrower two-point function, consistent with the CfA results. What seems to have happened is that the SGP field has intersected some abnormally rich clumps, which are abnormally wide in redshift. We cannot say what fraction of the width is due to peculiar velocities, but for our present purposes this does not matter. Now, using a field which intersects abnormally rich clumps will mean that the $k \to 0$ power will be higher than one would normally expect (although with our definition this may be offset by the increase in the normalizing factor of $1/N_{\rm gals}$ in the Fourier transform), and the extra width of the richer clumps will force the power spectrum to fall more rapidly with k. This combination of circumstances can clearly conspire to cause one to overestimate the significance of peaks at low k.

Unfortunately the north Galactic pole (NGP) data are as yet unpublished, so we were unable to perform the same kind of analysis here. However, (1) the visual impression is that the width of the clumps in the north is rather similar to that in the south; (2) there are again other unused fields which appear to show less spectacular clustering; (3) the redshift error distribution may broaden the pair separation distribution significantly; indeed, Koo, Kron, & Szalay (1987) quote a precision of $\Delta z = 0.005$ (or $\Delta v = 1500$ km s^{-1}), which would fit with our assumed clump width if they are quoting a 2 σ error. In short, it appears that our estimate for the clump width is somewhat larger than one would have expected, but for the fields actually used something close to 800 km s^{-1} does seem to be appropriate.

The mean power in Figure 1 seems nicely consistent with our estimate of $\langle | f_k^2 | \rangle = 0.042$, confirming that the peak of $k \simeq 0.05h$ Mpc^{-1} is not surprisingly high. What *is* striking in Figure 1, however, is the lack of power at smaller k (until we strike the transform of the selection function). The visual impression of the BEKS data is arguably due more to this sub-Poissonian degree of large-scale uniformity than to the presence of excess power concentrated at a single wavenumber. These points are amplified in Figure 3, which shows a set of simulations of the data as described above, including the clump redshift dispersion. In line with the above discussion, we have chosen to highlight realizations with a large contrast between power above and below $k \simeq 0.03h$ Mpc^{-1}. Although results as striking as those of BEKS do not occur on every occasion, it is certainly possible to come across things with similar visual appeal (and equally strong power-spectrum peaks) a reasonable fraction of the time.

There are several conclusions to be drawn. Although we have not simulated the BEKS data exactly, it is clear that the finite width of the clumps means that the principal peak in the power spectrum is very likely to lie at $\lambda \gtrsim 50h^{-1}$ Mpc. Estimation of the significance level of such a peak by reference to the average power at large k is certainly unreliable, and will produce a probability for the spike in the power which is too small. Both this consideration and the statistics of BEKS clump sizes suggest that their peak has a significance of $\sim 1\%$ at best. Given that the data can be reproduced by a model of Poisson clumps, there would seem to be little compelling evidence for either periodicity or large-scale power.

3. ONE-DIMENSIONAL POWER SPECTRA FROM GALAXY CLUSTERING STATISTICS

We have seen in the previous section that the form of $P(k)$ for $k \gtrsim 0.1h$ Mpc^{-1} is sensitive to the width of the clumps in redshift, and this undermines attempts to estimate the significance of the spike at low k by reference to the power at higher frequencies. Nonetheless, the power at long wavelengths ($k < 0.1$, say) can still be estimated from the spatial correlation function $\xi(r)$ following the approach of BEKS; they actually calculated the fluctuations in $\xi(\Delta z)$ at large separations this way, but the analysis is easily adapted to treat the power spectrum. Here we will elaborate on their calculation (which assumed a uniform cylindrical survey volume) to include realistic angular and radial selection. This is a useful complement to the Poisson clump model, since it gives an independent test of whether the clump properties assumed there are compatible with the null hypothesis of "ordinary" galaxy clustering (although, as we have noted, there is some indication that the deep southern field contains richer than average clumps). It also allows one to incorporate the expected change in the clustering pattern with the width of the survey volume (and therefore with distance from the observer); the Poisson clump model assumes rather unrealistically that the clump richness distribution is the same at high and low z. This approach also provides a useful starting point for calculating the statistics of *fluctuations* about the mean power level and the frequency of spikes in the power spectrum. To calculate such quantities apparently requires higher order galaxy correlation functions, which are poorly known. However, provided that there are enough independent "clumps" in the survey, the Fourier coefficients f_k will become Gaussian-distributed, and one can, in this approximation, obtain practically useful results.

Because the full-blown treatment for a realistic survey geometry is a little involved, we first (§ 3.1) present the analysis for a simpler situation: that of a uniformly sampled, long, thin tube. This suffices to illustrate the principles involved, and also to get a rough idea at least of the mean power level, which can be compared with the results of § 2 and also with the real data. We later (§ 3.3) generalize the analysis to treat a survey in which galaxies are selected according to some arbitrary angular and radial selection functions.

Before commencing, we discuss a few points concerning notation. The most common Fourier transform convention is that used by Peebles (1980), in which space is taken to be periodic over some arbitrary large distance $\mathscr{L}$. The Fourier relations in n dimensions are thus

$$F(x) = \left(\frac{\mathscr{L}}{2\pi}\right)^n \int \tilde{F}(k) \exp(-i\boldsymbol{k} \cdot \boldsymbol{x}) d^n k ,$$

$$\tilde{F}(k) = \left(\frac{1}{\mathscr{L}}\right)^n \int F(x) \exp(i\boldsymbol{k} \cdot \boldsymbol{x}) d^n x . \tag{3.1}$$

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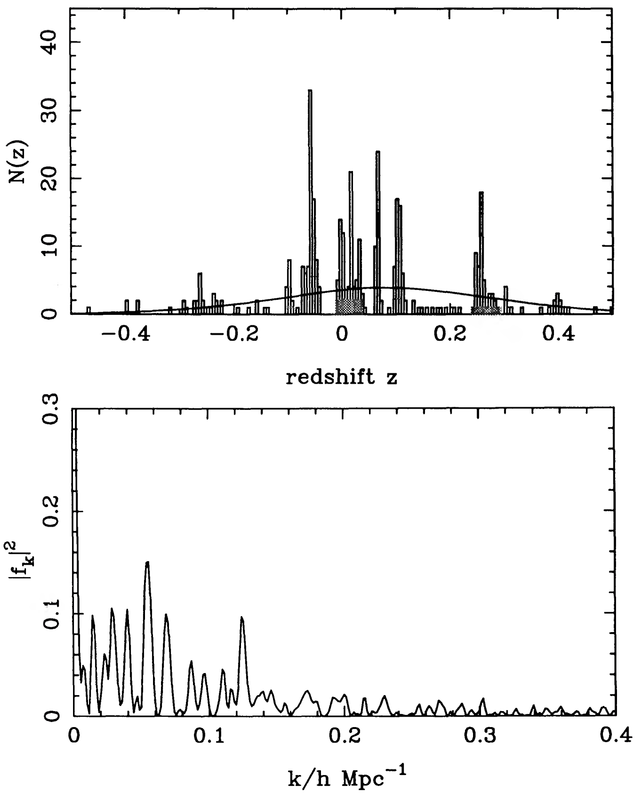

FIG. 3a

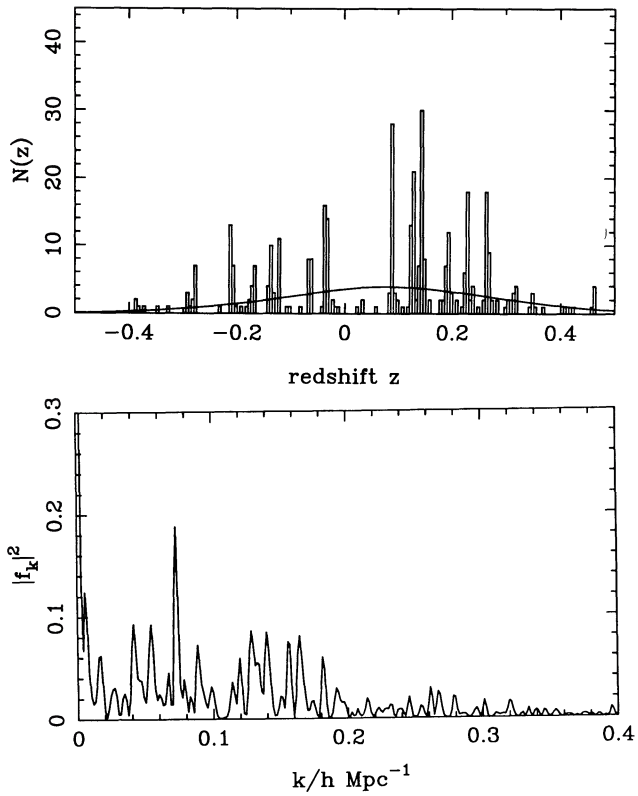

FIG. 3b

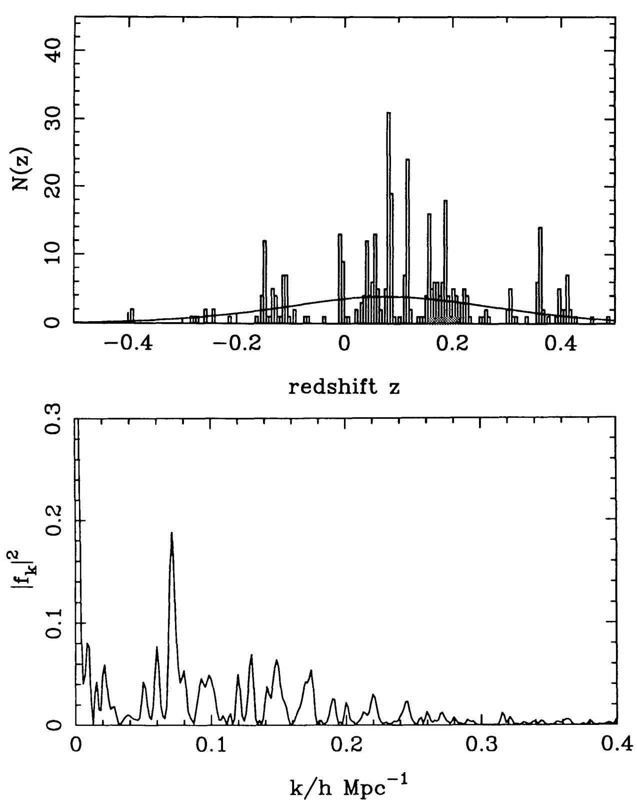

FIG. 3c

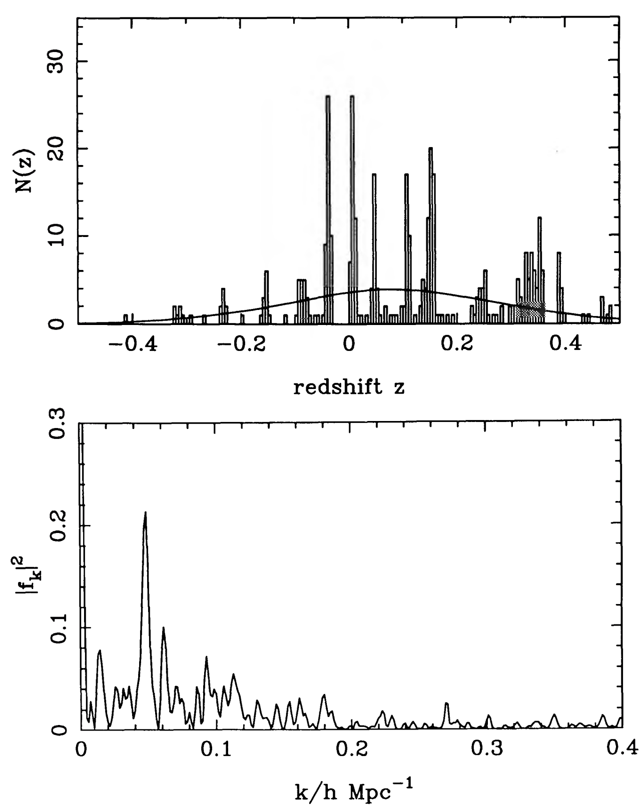

FIG. 3d

FIG. 3.—Catalog of simulated skewers together with their power spectra. We show the "best" results from runs of size (*a*) 1, (*b*) 10, (*c*) 30, and (*d*) 100. We selected realizations which not only displayed a high peak in the power but which also had abnormally low power at very low *k*. Had we selected only on peak height, significantly higher peaks would have appeared here. Even given the a posteriori nature of the selection criteria, there appears to be no difficulty in matching the BEKS data $\sim 1\%$ of the time.

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The advantage of this convention is that the presence of the density of states means that in the discrete case the integral goes over to a direct Fourier series: $F(x) = \Sigma \tilde{F}(k) \exp(-i\boldsymbol{k} \cdot \boldsymbol{x})$. However, since $\mathscr{L}$ never appears in the formulae for any observable quantity, we set $\mathscr{L} = 1$ in the analysis without loss of generality. As a further simplification of notation, we shall generally omit modulus signs, so for complex quantities $F^2 \equiv FF^*$. Finally, except where it is obvious from the context, z generally refers not to redshift but to the Cartesian comoving coordinate in the redshift direction.

3.1. *Idealized Survey Geometry*

If we have a statistically isotropic three-dimensional random field $\delta(\boldsymbol{r})$ with power spectrum $P(k) \equiv \langle \delta_k^2 \rangle$, but view this only along a line (the z-axis, say), the power spectrum of this one-dimensional field $F(z) = \delta(0, 0, z)$ is (e.g., Lumsden, Heavens, & Peacock 1989)

$$\langle \tilde{F}_k^2 \rangle = \frac{1}{(2\pi)^2} \int dk_x \int dk_y\, P(k_x, k_y, k) = \frac{1}{2\pi} \int_k^\infty P(y) y\, dy\,, \tag{3.2}$$

or, in terms of the variances per logarithmic interval of wavenumber $\Delta_{3\mathrm{D}}^2 \equiv d\sigma_{3\mathrm{D}}^2/d\ln k = k^3 P(k)/2\pi^2$, $\Delta_{1\mathrm{D}}^2 \equiv d\sigma_{1\mathrm{D}}^2/d\ln k = k\langle \hat{F}_k^2 \rangle/\pi$:

$$\Delta_{1\mathrm{D}}^2(k) = k \int_k^\infty \Delta_{3\mathrm{D}}^2(y) y^{-2}\, dy\,. \tag{3.3}$$

(Note the factor of 2 from $\pm k$ in one dimension.) So, provided that the three-dimensional power spectrum has an effective index $n > -2$ as $k \to 0$, and assuming that the integral converges at large k, the long-wavelength terms in the one-dimensional power spectrum are dominated by power projected from short wavelengths in three-dimensions, and are highly insensitive to the true three-dimensional power on large scales for most spectra.

This projection of spurious large-scale power is an inevitable effect in lower-dimensional redshift surveys and will always weaken any conclusions which can be drawn concerning large-scale density perturbations. We note in passing that the effect for a two-dimensional slice is similar to that in one dimension:

$$\Delta_{2\mathrm{D}}^2 = k^2 \int_k^\infty \Delta_{3\mathrm{D}}^2(y) \frac{y^{-2}\, dy}{\sqrt{y^2 - k^2}}\,. \tag{3.4}$$

Again, an effective $n = 0$ portion is produced at low k. Since Δ_{3D}^2 behaves very nearly as k^2, we see that $\Delta_{2D}^2/\Delta_{3\mathrm{D}}^2 \simeq \cosh^{-1}(k_c/k)$, where the integral is cut off at k_c by the slice thickness; this power ratio will be a factor of 3–5 in practice. Such enhancement of large-scale power needs to be kept in mind when interpreting surveys such as the CfA slice: the linear scale for unit variance in the projected areal density of galaxies will be rather higher than the corresponding three-dimensional length.

Galaxies are known to be clustered on small scales ($< 10h^{-1}$ Mpc or so) with a two-point function which is well described by a power law: $\xi \propto r^{-\gamma}$, with $\gamma \simeq 1.8$. Since this power-law behavior extends over a large range of length scales, the power spectrum should also be a power law (at least over the corresponding range of spatial frequencies) with index $n = \gamma - 3 \simeq -1.2$. Using this in the above formula, one finds that the one-dimensional power diverges. This is an artifact of the idealized survey geometry (an infinitely long, infinitesimally thin skewer).

A realistic survey will be finite in both length and breadth. As a next step toward realism, let us assume that we have a sample of the δ field confined to a cylinder of length L and radius R, and define

$$f_k(\boldsymbol{r}_0) = \int d^3r\, \delta(\boldsymbol{r}_0 + \boldsymbol{r}) w(z) W(x, y) \exp(ikz)\,, \tag{3.5}$$

where we have introduced the "window functions" $w(z)$, $W(x, y)$ which are assumed to be normalized so that $1 = \int dz\, w(z) = \iint dx\, dy\, W(x, y)$. The Fourier amplitude f_k is therefore a convolution of the δ field with a function which is the product of $w(z)W(x, y)$ with sinusoids of spatial frequency k. By the convolution theorem, the Fourier transform of $f_k(\boldsymbol{r}_0)$—a function of the position of the box $\boldsymbol{r}_0$—is the product of the Fourier transform of δ with a "response function" $\tilde{g}_k = \tilde{w}(k_z - k)\tilde{W}(k_x, k_y)$; the convolution of the Fourier transform of wW with a delta function at $\boldsymbol{k} = \{0, 0, k\}$, so

$$f_k(\boldsymbol{r}_0) = \frac{1}{(2\pi)^3} \int d^3k\, \tilde{\delta}(\boldsymbol{k}) \tilde{w}(k_z - k) \tilde{W}(k_x, k_y) \exp(i\boldsymbol{k} \cdot \boldsymbol{r}_0)\,. \tag{3.6}$$

There are three distinct scales in this problem: L, R, and $1/k$. We are interested in the regime where $L \sim H^{-1} \gg 1/k \gg R$. Thus, if we think of this experiment as an antenna for density perturbations, the response pattern is highly directional, being very extended in the k_x, k_y directions, with width $\sim 1/R$, while along the line of sight $\tilde{g}$ is very narrow, with width $\delta k_z \sim 1/L$ about the plane $k_z = k$. For $k \gg 1/R$ the response function cuts off, and the divergent contribution from high wavenumber is avoided. For the $k^{-1.2}$ power-law power spectrum, the main contribution then comes from wavenumbers $\sim 1/R$. Note that, provided that $k \ll 1/R$, the one-dimensional power spectrum is independent of k.

The power spectrum is

$$\langle f_k^2 \rangle = \frac{1}{(2\pi)^3} \int d^3k\, \tilde{w}^2(k_z - k) \tilde{W}^2(k_x, k_y) P(k)\,. \tag{3.7}$$

For large L, $\tilde{w}^2(k) \to 2\pi\, \delta_D(k)/L$, where δ_D denotes the Dirac delta function, and we have

$$\langle f_k^2 \rangle = \frac{1}{(2\pi)^2} \int d^2k\, \tilde{W}^2(k) P(\sqrt{k_z^2 + k^2})\,. \tag{3.8}$$

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The analog of equation (3.3) is

$$\Delta^2_{1\mathrm{D}}(k) = \frac{Lk\langle f_k^2\rangle}{\pi} = k\int_k^\infty \Delta^2_{3\mathrm{D}}(y)\tilde{W}^2(\sqrt{y^2-k^2})y^{-2}\,dy\,. \tag{3.9}$$

As a concrete example, suppose (as did BEKS) that the transverse selection function is a disk of radius R. In this case,

$$\tilde{W}(k) = \frac{2}{kR}J_1(kR) \simeq 1 - \frac{(kR)^2}{8} + \cdots \tag{3.10}$$

and J_1 is a Bessel function. To second order in k, we can thus approximate this by the Gaussian $W = \exp[-(kR)^2/8]$, which is analytically convenient in performing the integration for $\Delta^2_{1\mathrm{D}}$ (see below). With a pure power-law correlation function, $\xi(r) = (r/r_0)^{-\gamma}$, the corresponding three-dimensional power spectrum is

$$\Delta^2_{3\mathrm{D}} = \frac{2}{\pi}(kr_0)^\gamma\Gamma(2-\gamma)\sin\frac{(2-\gamma)\pi}{2} \equiv \beta(kr_0)^\gamma \tag{3.11}$$

[$= 0.903(kr_0)^{1.8}$ if $\gamma = 1.8$]. The one-dimensional power spectrum with the Gaussian window approximation is

$$\Delta^2_{1\mathrm{D}}(k) = \beta kR\left(\frac{r_0}{R}\right)^\gamma \exp\left[\frac{(kR)^2}{4}\right]\int_{kR}^\infty x^{\gamma-2}\exp\left(-\frac{x^2}{4}\right)dx\,. \tag{3.12}$$

As $k \to 0$, this becomes

$$\Delta^2_{1\mathrm{D}}(k) = \beta kR\left(\frac{r_0}{R}\right)^\gamma 2^{\gamma-2}\Gamma\left(\frac{\gamma-1}{2}\right) \tag{3.13}$$

[$= 1.74kR(r_0/R)^{1.8}$ for $\gamma = 1.8$].

It is easily seen that this is essentially equivalent in content to the result for Poisson clumps: $\langle f_k^2\rangle = 1/N_{\mathrm{clumps}}$: If the mean number density of galaxies is n, then the typical number of neighbors within distance R is $\sim n\xi(R)R^3$. If we interpret this as the typical multiplicity for clumps found in a tube of radius R, the average number of clumps found will be $N_{\mathrm{clumps}} \sim nLR^2/n\xi(R)R^3$, so $\Delta^2_{1\mathrm{D}} \sim Lk\langle f_k^2\rangle \sim kR\xi(R)$, in accord with equation (3.13). Note that we are ignoring the contribution to the power from the Poisson effects arising from the discreteness of individual galaxies. This could easily be incorporated here and in what follows, but it turns out to be a relatively small effect and so we have omitted it for the sake of clarity.

We can use the above to obtain an approximate *external* prediction of the mean power. If we take $r_0 = 5h^{-1}$ Mpc and the BEKS assumption of $R = 3h^{-1}$ Mpc, we get $\Delta^2_{1\mathrm{D}} = 0.64$ for $\lambda = 128h^{-1}$ Mpc (cf. $\Delta^2_{3\mathrm{D}} = 0.072$ at this point). Note the enormous increase in $\Delta^2_{1\mathrm{D}}$ owing to aliased small-scale power. It is immediately clear that the sort of extra large-scale power which is needed to show up above this background is going to be vast: order-unity density fluctuations on scales ~ 100 Mpc. Note that equation (3.12) predicts a fall in $\Delta^2_{1\mathrm{D}}$ from the $k = 0$ value by a factor of 1.4 at $k = 0.2h\ \mathrm{Mpc}^{-1}$ and 2 at $k = 0.4$. Although smaller than the correction factors found in § 2, this is a further reminder that, even in the absence of random velocities and redshift errors, the "white noise" plateau in $\Delta^2_{1\mathrm{D}}$ is of strictly limited extent.

The expected aliased large-scale power for one mode is $f_k^2 = (\pi/kL)\Delta^2_{1\mathrm{D}} = 0.019$, or about 45% of the prediction of the Poisson clump model of § 2. The difference between these results, while fairly modest, has important implications. The peak in the power spectrum found by BEKS would certainly appear very significant if we compared it with the "baseline" derived above, and one would certainly be able to reject the "null hypothesis" that there is power arising from small scales with this level. Indeed, this is precisely what BEKS concluded, having performed the equivalent calculation on the pair counts in their sample. However, this still leaves two possible interpretations: one can conclude either that there is evidence for large-scale power of some kind or that the calculation of the power level implied by the null hypothesis has not been performed accurately enough. The Poisson clump model suggests that the latter explanation is the correct one, and there are various reasons that make this plausible. One possible reason for the discrepancy is that the galaxy correlation function we have adopted is, for some reason, not appropriate for the galaxies found by BEKS. The raw power spectrum of the counts is very sensitive to the appropriate value of ξ, which, as discussed above, may well be larger than average for the selected fields. However, with our definition of f_k^2, the normalizing factor of $1/N^2$ will tend to cancel this out. For a given spatial distribution of clumps, our definition of the power is unchanged if the multiplicity of all clumps is changed by some factor. While it is difficult to guess how the number of clumps might have been affected by selection, we suspect that this is a weaker effect. Another potential source of error is our very idealized modeling of the geometry of the sample. The redshift distribution is far from uniform over $z = \pm 0.5$; with our definition the expected power $\langle f_k^2\rangle$ varies inversely as the length of the tube, so this could be a big effect. Since the significance of any spikes in the power spectrum depend sensitively on what one assumes for the mean power level, it is important to incorporate these features in the analysis. This more general analysis is performed in § 3.4. First, however, we will explore some further properties of power spectra using this simplified model.

3.2. *Power Spectra in the Central Limit*

Provided that there are enough independent clumps, the real and imaginary parts of f_k [denoted by $c(k)$ and $s(k)$, respectively] will become Gaussian-distributed random variates, and so the distribution of these fields and their derivatives is entirely specified by the correlation coefficients $\langle c^2(k)\rangle$, $\langle c(k)s(k)\rangle$, etc. This result is a consequence of the central-limit theorem and *does not* assume anything about the statistics of the small-scale clustering; see § 3.6 for further discussion. In particular, we are not assuming that the small-scale structure is Gaussian; it is clearly not so. All of the needed two-point functions can be calculated from equation (3.5),

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more or less as we did for $\langle f_k^2 \rangle = \langle c(k)^2 \rangle + \langle s(k)^2 \rangle$. In the present case, where the window function factorizes into separate transverse and longitudinal functions, we can simply factor out the dependence of the transverse k components to give, for the c field, for instance,

$$c(k) = \frac{1}{2\pi} \int dk_z \Delta(k_z) \int dz\, w(z) \cos(kz) \exp(ik_z z) , \tag{3.14}$$

where

$$\Delta(k_z) \equiv \frac{1}{(2\pi)^2} \iint dk_x dk_y \tilde{W}(\sqrt{k_x^2 + k_y^2}) \delta(k_x, k_y, k_z) \exp(i\boldsymbol{k} \cdot \boldsymbol{r}_0) \tag{3.15}$$

is a complex random variable with $\langle \Delta(k_z) \Delta(-k_z') \rangle = \langle \Delta^2 \rangle \delta_D(k_z - k_z')$, and

$$\langle \Delta^2 \rangle = \frac{1}{(2\pi)^2} \int d^2k\, \tilde{W}^2(k) P(\sqrt{k^2 + k_z^2}) = 2\pi \langle f_{k_z}^2 \rangle . \tag{3.16}$$

We have invoked here the assumptions that $\delta(\boldsymbol{r})$ is a real and statistically homogeneous process (though we have not assumed that it is a Gaussian process) to set $\langle \delta(\boldsymbol{k}) \delta^*(\boldsymbol{k}') \rangle = (2\pi)^3 P(k) \delta_D(\boldsymbol{k} - \boldsymbol{k}')$. For the modes with k_x, $k_y \sim 1/R$ which dominate, $\langle \Delta^2 \rangle$ is independent of k_z, so we have

$$\begin{aligned}
\langle c(k)c(k') \rangle &= \frac{\langle f_k^2 \rangle}{2\pi} \int dz\, w(z) \cos(kz) \int dz' w(z') \cos(k'z') \int dk \exp[ik(z + z')] \\
&= \langle f_k^2 \rangle \int dz\, w^2(z) \cos(kz) \cos(k'z') \\
&= \frac{\langle f_k^2 \rangle}{2} \int dz\, w^2(z) \{\cos[(k - k')z] + \cos[(k + k')z]\} .
\end{aligned} \tag{3.17}$$

For $k, k' \gg 1/L$, the second term here will always be small, since $\cos[(k + k')z]$ will oscillate many times within the boxcar. The first term, however, can be large, provided that $k - k'$ is smaller than or comparable to $1/L$, so we have in this approximation

$$\langle c(k)c(k') \rangle = \frac{\langle f_k^2 \rangle}{2} \int dz\, w^2(z) \cos[(k - k')z] , \tag{3.18}$$

which is a function only of $\delta k \equiv k - k'$, so $c(k)$ is a *stationary* Gaussian process. In the same limit, an identical expression is found for $\langle s(k)s(k') \rangle$, while

$$\langle s(k)c(k') \rangle = \frac{\langle f_k^2 \rangle}{2} \int dz\, w^2(z) \sin[(k - k')z] . \tag{3.19}$$

The covariance matrix elements involving the derivatives of c and s are obtained quite simply. We have, for instance,

$$c'(k) \equiv \frac{dc(k)}{dk} = -\frac{1}{2\pi} \int dk_z \Delta(k_z) \int dz\, zw(z) \sin(kz) \exp(ik_z z) , \tag{3.20}$$

and consequently, for example,

$$\langle s(k)c'(k') \rangle = \frac{\langle f_k^2 \rangle}{2} \int dz\, zw^2(z) \cos[(k - k')z] . \tag{3.21}$$

It is interesting that these covariance functions "know" about the origin of our spatial coordinates. If the window function $w(z)$ is symmetric, and we take the origin to lie at the center of symmetry, then the c and s fields are completely independent. If we place the origin elsewhere, this clearly has no effect on the power spectrum, but now the real and imaginary parts are correlated. What happens here is that if we shift the origin by Δz, this has the effect of rotating f_k in the Argand plane by an angle $k\,\Delta z$. This spiraling about the k-axis introduces correlations between even derivatives of c and odd derivatives of s and vice versa.

Armed with these correlation functions, we can calculate the frequency of peaks in the power spectrum. Another useful quantity we shall extract is the variance in the mean power estimated from a finite sample. One way to calculate this is via a model for c and s:

$$\begin{aligned}
c(k) &= \int dz\, w(z) g(z) \cos(kz) , \\
s(k) &= \int dz\, w(z) g(z) \sin(kz) ,
\end{aligned} \tag{3.22}$$

where $g(z)$ is a real, stationary, zero-mean, Gaussian random field with $\langle g(z)g(z') \rangle = \langle f_k^2(z) \rangle \delta_D(z - z')$. Equivalent results can also be obtained with a representation of the field as a set of independent shots with constant strength, and with number density proportional to $w^2(z)$. Whichever approach is taken, the resulting correlation matrix elements are identical to those calculated

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above, so provided the central limit is valid, the statistical properties of the power spectrum obtained in this way should be correct, independent of approach. We therefore introduce a general concept to describe this situation. Quantities such as the correlation functions of c and s can be calculated as integrals over something we shall refer to as the spatial fluctuations *intensity* (in other applications, one would call this the power spectrum, but we clearly wish to reserve that name here for $c^2 + s^2$):

$$\langle f_k f_{k+\Delta k}\rangle \equiv \langle f_k^2\rangle\psi(\Delta k) = \int dx\, \mathscr{I}(x) \exp(i\Delta kx)\,,$$

$$\langle c(k)c(k+\Delta k)\rangle \equiv \frac{\langle f_k^2\rangle}{2}\psi_+(\Delta k) = \frac{1}{2}\int dx\, \mathscr{I}(x)\cos(\Delta kx)\,, \tag{3.23}$$

$$\langle c(k)s(k+\Delta k)\rangle \equiv \frac{\langle f_k^2\rangle}{2}\psi_-(\Delta k) = \frac{1}{2}\int dx\, \mathscr{I}(x)\sin(\Delta kx)\,,$$

where $\mathscr{I}$ denotes the intensity.

We are now in a position to construct realizations of power spectra with the correct correlation properties. Before doing this, however, we will first generalize the calculation to a realistic survey geometry.

3.3. *Realistic Survey Geometry*

We want to consider now an observer sitting at $\boldsymbol{r}_0$ who takes redshifts along a narrow cone parallel to the z-axis. We will assume that the galaxies are selected in some way on their apparent magnitude or size, resulting in some radial selection function $\phi(z)$, and on angle, which we can describe in terms of an angular selection function $W(\theta_x, \theta_y)$, this function being the probability that a galaxy at that position on the sky is included in the survey (assuming, of course, that the galaxy meets the magnitude selection criterion).

This observer takes a one-dimensional Fourier transform of the counts:

$$f_k(\boldsymbol{r}_0) = c(k) + is(k) = \frac{1}{N_{\rm gals}}\int dz\, n(z)\exp(ikz)\,. \tag{3.24}$$

We show in Appendix A that, for a three-power spectrum $P = P_0 k^n$, the covariance matrix elements for c and s can be expressed in a very similar form to equations (3.23):

$$\langle c(k)c(k')\rangle = \langle s(k)s(k')\rangle = \frac{\langle f_k^2\rangle}{2}\psi_+(k-k') \equiv \frac{\langle f_k^2\rangle}{2}\int dz\, w^2(z)\cos[(k-k')z]\,, \tag{3.25}$$

$$\langle c(k)s(k')\rangle = \frac{\langle f_k^2\rangle}{2}\psi_-(k-k') \equiv \frac{\langle f_k^2\rangle}{2}\int dz\, w^2(z)\sin[(k-k')z]\,, \tag{3.26}$$

where

$$\langle f_k^2\rangle \equiv \frac{P_0}{2\pi}\frac{\int dz\, z^{2-n}\phi^2(z)}{[\int dz\, z^2\phi(z)]^2}\frac{\int d\kappa\, \kappa^{1+n}\tilde{W}^2(\kappa)}{[\int d^2\theta\, W(\theta)]^2}\,, \tag{3.27}$$

and where now

$$w(z) \equiv z^{1-n/2}\phi(z)\Big/\left[\int dz\, z^{2-n}\phi^2(z)\right]^{1/2}\,. \tag{3.28}$$

As before, these results are derived under the assumption that the power is dominated by modes with $(k_x^2 + k_y^2)^{1/2} \sim H/\theta \gg k, k'$, and that k and k' are themselves very much greater than H. Note that, in contrast to the cylindrical case, the spectral index for the small-scale clustering enters. This is because the "color" of the power spectrum determines which range of distances from the observer tend to dominate the one-dimensional power: If the spectrum is very blue (positive n), the dominant contribution comes from relatively nearby, where the width of the cone is small, while for redder spectra the contribution to the power is weighted more toward large distances. Thus, the spatial fluctuation intensity $\mathscr{I}(z) \propto w^2(z)$ depends both on the selection function ϕ and on the power spectrum. Another difference is that while, with the symmetric boxcar "selection function" $w(z)$ we could choose the spatial origin so that the c and s fields were independent, this is no longer possible. Just as before, a model for $c(k)$, $s(k)$ in which these are Gaussian-distributed and have the desired two-point correlation functions (3.25)–(3.28) is given by equation (3.22).

We also show in Appendix A that the contribution to the one-dimensional power spectrum from modes with wavenumber very much less than H/θ is

$$\langle f_k^2\rangle = \frac{1}{2\pi}\frac{\int dz\, z^4\phi^2(z)}{[\int dz\, z^2\phi(z)]^2}\int_k dk\, kP(k)\,. \tag{3.29}$$

This is used in § 4.1, where we discuss the level of extra large-scale power which might be detectable in the BEKS survey. Finally, recalling from equation (3.2) that the one-dimensional power spectrum is $P_{\rm 1D} = (2\pi)^{-1}\int dk\, kP(k)$, we can form an estimator of any intrinsic long-wavelength power:

$$\hat{P}_{\rm 1D}(k) = \frac{[\int dz\, z^2\phi(z)]^2}{\int dz\, z^4\phi^2(z)}\left\{f_k^2 - \frac{P_0}{2\pi}\frac{\int dz\, z^{2-n}\phi^2(z)}{[\int dz\, z^2\phi(z)]^2}\frac{\int d\kappa\, \kappa^{1+n}\tilde{W}^2(\kappa)}{[\int d^2\theta\, W(\theta)]^2}\right\}\,. \tag{3.30}$$

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3.4. *Peaks in Power Spectra*

Armed with the covariance matrix for the c and s fields, one can calculate various properties of peaks in power spectra; the most useful are the abundance and height distribution, since these allow us to assign exact significance levels to observed peaks. The details of the calculation are given in Appendix B, but the main results are as follows. First define some dimensionless variables in terms of power ($f_k^2 = c^2 + s^2$) and the moments about the mean over the intensity [$\sigma_m^2 \equiv \int \mathscr{I}(x)(x - \bar{x})^{2m} dx$, where $\bar{x} \equiv \int \mathscr{I}(x)x\, dx / \int \mathscr{I}(x) dx$]:

$$\nu \equiv f_k/\sigma_0 \, , \qquad \gamma \equiv \sigma_1^2/(\sigma_0 \sigma_2) \, , \qquad \epsilon \equiv \sigma_{3/2}^2/(\sigma_1 \sigma_2) \, , \qquad k_* \equiv \sigma_1/\sigma_2 \, . \tag{3.31}$$

To translate results to power terms, note that $\langle P \rangle = \langle f_k^2 \rangle = 2\sigma_0^2$, so that

$$\frac{P}{\langle P \rangle} = \frac{\nu^2}{2} \, . \tag{3.32}$$

In these terms, the differential peak density is

$$\frac{\partial N}{\partial \nu} = \frac{\nu}{(2\pi)^{3/2} k_*} \exp\left(-\frac{\nu^2}{2}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{y^2}{2}\right) dy \left\{ a \exp\left(-\frac{b^2}{2a^2}\right) + b \left(\frac{\pi}{2}\right)^{1/2} \left[1 + \mathrm{erf}\left(\frac{b}{\sqrt{2}\, a}\right)\right] \right\} , \tag{3.33}$$

where $a \equiv (1 - \gamma^2 - \epsilon^2)^{1/2}$ and $b \equiv \gamma\nu + \epsilon y - \gamma y^2/\nu$.

The parameter ϵ describes the asymmetry of the selection function; we show in Appendix B that for all practical purposes one can proceed by assuming it to be zero, in which case an excellent approximation for the total density is $N = (1 + \gamma^{1.44})/(2\pi k_*)$. For even moderately high peaks, we also show that the high-peak integral density is an excellent approximation (provided that $\gamma \gtrsim 0.3$):

$$N(>\nu) = \frac{\gamma}{(2\pi)^{1/2} k_*} \nu \exp\left(-\frac{\nu^2}{2}\right) , \tag{3.34}$$

and we therefore use this in what follows. As usual, this shows that high peaks are not as improbable as one might naively expect; for $\gamma = 1$, 1% of power-spectrum peaks have $\nu > 3.5$ (i.e., $P/\langle P \rangle > 6$, for which the field-point probability is 2×10^{-3}).

The parameter γ describes the degree of concentration of the intensity. For example, in the case of the uniform tube, γ is just the ratio $\langle x^2 \rangle / \langle x^4 \rangle^{1/2}$ for a top hat, which is $\gamma = 5^{1/2}/3 = 0.745$.

3.5. *Uncertainty in the Mean Power from a Finite Bandwidth*

If we make an estimate the mean power over some band of wavenumber,

$$\hat{P} = \frac{1}{\Delta k} \int_k^{k+\Delta k} dk\, f_k^2 \, , \tag{3.35}$$

the expectation value is just $\langle \hat{P} \rangle = \langle f_k^2 \rangle$, and, using the model (3.22), (3.28) for f_k, the second moment is

$$\begin{aligned} \langle \hat{P}^2 \rangle &= \frac{1}{(\Delta k)^2} \int_k^{k+\Delta k} dk \int_k^{k+\Delta k} dk' \int dz \int dz' \int dz'' \int dz''' \, w(z)w(z')w(z'')w(z''') \langle g(z)g(z')g(z'')g(z''') \rangle \exp\,[ik(z - z')] \exp\,[ik'(z'' - z''')] \\ &= \langle \hat{P} \rangle^2 + \frac{\langle f_k^2 \rangle^2}{(\Delta k)^2} \int_k^{k+\Delta k} dk \int_k^{k+\Delta k} dk' [\psi^2(k - k') + \psi^2(k + k')] \, . \end{aligned} \tag{3.36}$$

For $k, k' \gg H$ we can neglect the second term, and the variance in $\hat{P}$ is then independent of k. This has the reasonable properties that for $\Delta k \to 0$, $\langle \hat{P}^2 \rangle - \langle \hat{P} \rangle^2 = \langle \hat{P} \rangle^2$ as expected for a single sample of an exponentially distributed variable, whereas for $\Delta k \gg k_c$, where k_c is a coherence scale [roughly the width of the $\psi(k)$, which will be on the order of H], $\langle \hat{P}^2 \rangle - \langle \hat{P} \rangle^2 \simeq (k_c/\Delta k) \langle \hat{P} \rangle^2$. Thus, the rms fractional fluctuation in $\hat{P}$ will be on the order of $(\Delta k/k_c)^{-1/2}$, i.e., roughly the inverse root of the number of "coherence lengths" in the sample. This can be made precise:

$$\sigma_{\hat{P}}^2 \equiv \frac{\langle \hat{P}^2 \rangle - \langle \hat{P} \rangle^2}{\langle \hat{P} \rangle^2} = \frac{1}{\Delta k} \int_{-\infty}^{\infty} dk\, \psi^2(k) \equiv \frac{k_c}{\Delta k} \, , \tag{3.37}$$

where Parseval's theorem gives

$$k_c = 2\pi \int dr\, \mathscr{I}^2(r) \Big/ \left[\int dr\, \mathscr{I}(r) \right]^2 . \tag{3.38}$$

3.6. *Applicability of the Central-Limit Theorem*

The foregoing results are exact in the limit that the number of independent fluctuations in the data gives to infinity. In this limit, the power spectrum loses all knowledge of any high-order correlation functions of the small-scale density field. Realistically (and especially for the BEKS survey) the power spectrum will be dominated by a modest number of independent clumps, so one might question how valid the central-limit result is. One indication that the central limit does apply quite well is that with the Poisson clump model of § 2 we found that the second moment of the power was within about 5% of the central-limit prediction. To explore further the consequences of departures from the central limit, we have made a series of numerical experiments, laying down a set of

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N equal weight shots uniformly within a boxcar and calculating the power spectrum. We found that even with $N = 5$ or 6, the power spectrum looks very much like the central-limit prediction, and with 10 shots the result is practically indistinguishable from the $N \to \infty$ limit. The central limit must go wrong for extreme fluctuations, since for finite N the maximum power is N^2. For $N = 7$, for instance, the central-limit result predicts that the power should exceed this $\simeq 10^{-3}$ of the time, so, if we are satisfied with predictions accurate to the $\simeq 1\%$ level, N can be very small indeed.

4. APPLICATION TO THE BEKS SURVEY

We have already (§ 3.1) estimated the mean power for the whole BEKS survey using the uniform cylinder model; this yielded $\langle f_k^2 \rangle \simeq 0.019$, as against the observed peak value of 0.21. The implied probability of $p = \exp(-0.21/0.019) = 10^{-4.8}$ seems to provide convincing rejection of the null hypothesis. However, we can now use our results on peaks to allow for the fact that BEKS chose the highest peak in the "undamped" range, $\Delta k = 0.1h$ Mpc^{-1}. The necessary moments are $k_* = 1.2 \times 10^{-3} h$ Mpc^{-1} and $\gamma = 0.75$ for a uniform cylinder. The expected number of peaks with ν above the observed value of 4.7 in $\Delta k = 0.1$ is then $10^{-2.7}$. To raise this number to unity requires a mean power level only a factor of ~ 1.5 times higher than our crude estimate, and we therefore need to compare with the results of our more exact modeling.

The data used in BEKS are an amalgam of various surveys with heterogeneous selection functions. The new data are the very narrow deep surveys in the SGP and NGP. To these were added some shallower, but much wider angle, surveys which fill out the picture at low redshift. While our assumption that the selection function factorizes into an angular and radial component is clearly not applicable to the combined survey, we can treat these three components separately and then add their contributions to $\langle f_k^2 \rangle$, and to the correlation functions (weighted by the square of the number of galaxies).

The radial selection functions can be estimated from the histogram of counts. We fitted these to the functional form $dn/dr = r^2 \phi(r)$ with $\phi(r) = \phi_* r^{-1/2} \exp(-r/r_*)^2$. This function accounts quite well for the distribution in depth in all cases. For the NGP we found $r_* = 551 h^{-1}$ Mpc, for the SGP $r_* = 536 h^{-1}$ Mpc, and for the shallower surveys $r_* = 134 h^{-1}$ Mpc. The spatial fluctuation amplitudes $\mathscr{I}^{1/2} \propto w(z)$ for the three components are shown in Figure 4. Also shown is the combination obtained by adding the separate components in quadrature. We should mention here that in § 3 we implicitly assumed that the normalizing factor $1/N$ in our definition of f_k is known accurately in advance, whereas one has to estimate this from the actual data. In an ideal world this would not be a problem, but with realistic surveys there may be appreciable fluctuations in the number of galaxies. With our approach it is very difficult to deal with this problem, or with the related difficulty of possibly having fields with richer than average clumps, and we will simply ignore it. If it turns out that any such bias leaves the effective number of clumps unchanged (i.e., in the richer fields there tend to be richer clusters), then our approach is justified. If not, we need better data.

The angular selection function is somewhat trickier. We begin with the NGP survey. Apparently (D. C. Koo 1990, personal communication) the survey was confined to a cone of radius 20′, but this was not uniformly sampled; rather, a small fraction of the angle was surveyed in roughly circular patches of radius $\simeq 2'.5$. Our model for the angular selection function here is that we lay down a disk of radius $\theta_1 = 20'$, and within that disk scatter down small disks of radius $\theta_2 = 2'.5$ at random and with density such that the mean number of disks is N. If we define $\tilde{W}_\theta(\kappa)$ to be the Fourier transform of a uniform disk of size θ, we have

$$\tilde{W}_\theta(\kappa) = \pi \theta^2 h(\kappa\theta) , \qquad h(y) \equiv \frac{2J_1(y)}{y} . \tag{4.1}$$

The window function for our randomly placed disks is a convolution of a point process with a smaller disk, so we have

$$\tilde{W}(\kappa) = \tilde{W}_1(\kappa) \tilde{W}_{\theta_2}(\kappa) , \tag{4.2}$$

where $\tilde{W}_1$ is the Fourier transform of the N randomly placed points. We can obtain the expectation value (i.e., the mean over many realizations of random points) for $\tilde{W}^2$ as follows: divide the disk up into microcells labeled by some index i, and let n_i be the

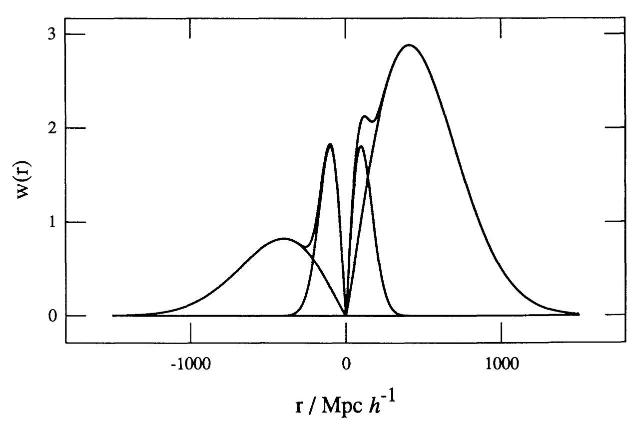

FIG. 4.—Plot of the spatial fluctuation amplitude $\mathscr{I}^{1/2}$ for the BEKS data. The intensity $\mathscr{I}$ is the quantity which, on application of a Fourier transform, yields the normalized correlation function in k-space for f_k. The different curves show the contributions of the NGP, SGP and shallow surveys from which the BEKS sample is constructed.

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occupation number for the ith cell. The moments of n are $\langle n \rangle = \langle n^2 \rangle = \cdots = N\, d^2 r/(\pi\theta_1^2)$, and $\langle n_i n_j \rangle = \langle n \rangle^2$ for $i \neq j$. Now

$$\tilde{W}_1(\kappa) = \sum_i n_i \exp(-i\kappa\theta_i)\,, \tag{4.3}$$

so

$$\begin{aligned}\langle \tilde{W}_1^2(\kappa) \rangle &= \sum_i \sum_j \langle n_i n_j \rangle \exp[i\kappa \cdot (\theta_i - \theta_j)] \\ &= \sum_i \langle n_i \rangle + \sum_i \sum_{j \neq i} \langle n_i n_j \rangle \exp[i\kappa \cdot (\theta_i - \theta_j)] \\ &= N + N^2 h^2(\kappa\theta_1)\end{aligned} \tag{4.4}$$

giving

$$\langle \tilde{W}^2(\kappa) \rangle = d\Omega^2 h_{\theta_2}^2(\kappa)[h_{\theta_1}^2(\kappa) + N^{-1}]\,. \tag{4.5}$$

The appropriate value for N is somewhat uncertain. Apparently there are about 10 zones in all, but some of these have yet to be completed. In the numerical calculations we have generally adopted $N = 5$, but have explored the consequences of varying N between 3 and 10, which should bracket the range of possibilities.

For the SGP, things are somewhat simpler: $W(\theta)$ is a filled cone of radius $\theta_3 = 10'$, so we have $\langle \tilde{W}^2(\kappa) \rangle = d\Omega^2 h_{\theta_3}^2(\kappa)$. For the shallow surveys the situation is again rather complicated. However, the counts seem to be dominated by four surveys, two in the north and two in the south, each of solid angle $d\Omega \simeq 14$ square degrees. We modeled these as pairs of disks of radius $\theta_4 = 126'$, and assumed that these were quite widely separated, so we have $\langle \tilde{W}^2(\kappa) \rangle = d\Omega^2 h_{\theta_4}^2(\kappa)/2$.

Together with the power spectrum $P(k)$, we now have all the ingredients required for a complete statistical description of the power spectrum. The coherence lengths k_*, k_c and also γ, ϵ, and $\langle f_k^2 \rangle$ for the three survey components are given in Table 2. Let us now compare the predictions with the spectra actually found.

Since the data for the north at least are not published, we had to work with the counts from the histogram of BEKS. These are binned in cells of $\delta v = 1500$ km s^{-1} width, and we must understand how this will affect the power spectrum derived from the reduced information at our disposal. This is fairly straightforward: the act of binning is mathematically equivalent to first convolving the true counts with a boxcar of width δv, and then sampling this on (i.e., multiplying by) a comb with teeth at separation δv. The Fourier transform of the binned counts (considered as a set of delta functions) is therefore the product of the true transform and the Fourier transform of the boxcar then convolved with the Fourier transform of the comb. This latter convolution simply replicates the spectrum at high frequencies, and has no effect on the frequencies we will use. The convolving with the boxcar, though, will reduce the power we estimate from the histogram below the true power level. At the location of the peak ($k \simeq 0.05$) the power is down by a negligible 5%. For $k = 0.1$, it is down by 20%, and the suppression increases very strongly at higher k. For the wavenumbers of interest here this is a small effect, and we can safely use the binned counts. In Figures 5*a* and 5*b* we show the counts and power spectrum for the NGP survey alone. The "periodicity" is certainly quite striking to the eye in the counts, and in the power spectrum one sees a bump at the corresponding wavenumber. The mean power predicted from small-scale clustering is $\langle f_k^2 \rangle = 0.105$, so the peak power is $\simeq 2.3$ times the mean prediction. The power spectrum drops beyond $k \simeq 0.1h$ Mpc^{-1} because of velocity smoothing and the finite width of the histogram bins. For a sample of bandwidth $\Delta k \simeq 0.1h$ Mpc^{-1}, the predicted frequency of such events is 0.90, so this peak is by no means out of the ordinary. A similar conclusion is obtained by comparing with Figures 6*a*–6*d*. Here we have generated a sample of length $\Delta k = 1$ of the power spectrum according to the model (3.22). For $k \gtrsim H$ this is a stationary process, so what we have is effectively 10 independent samples the size of the BEKS usable bandwidth. The texture of this field is very similar to that found in the data. This is not too surprising, as the coherence length is set by the shape of the radial selection function, and the less well determined angular selection function only enters in determining the mean power level. If anything, the actual sample seems to show slightly less power than predicted. However, for this sample $k_c \simeq 10^{-2}h$ Mpc^{-1}, so from equation (3.36) we would expect fractional fluctuations in the mean power of about 33%, and the slight deficit seen is not at all unreasonable.

In Figures 7*a* and 7*b* we show the counts and power spectrum for the NGP plus SGP deep surveys. The texture is now more jagged, and in Figures 8*a*–8*d* we see this tendency also in the random realizations. This is quite easily understood from elementary properties of Fourier transforms. The window function $w(z)$ for the NGP plus SGP surveys combined looks crudely like the convolution of the window for one survey with a pair of delta functions with a separation somewhat larger than the effective depth of a single survey. The width of a peak is associated with the depth for a single survey, and the finer scale jagged features are associated with the separation of the two delta functions.

TABLE 2

Moments for the BEKS Data

Survey	$\langle f_k^2 \rangle$	k_c (h Mpc^{-1})	k_* (h Mpc^{-1})	γ	ϵ
NGP	1.05×10^{-1}	9.60×10^{-3}	3.05×10^{-3}	0.568	0.265
NGP + SGP	6.66×10^{-2}	8.29×10^{-3}	1.32×10^{-3}	0.395	−0.580
Combined	3.15×10^{-2}	6.74×10^{-3}	1.54×10^{-3}	0.484	−0.435

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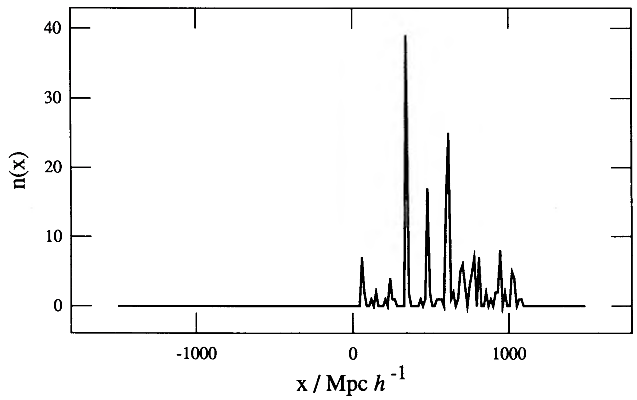

FIG. 5a

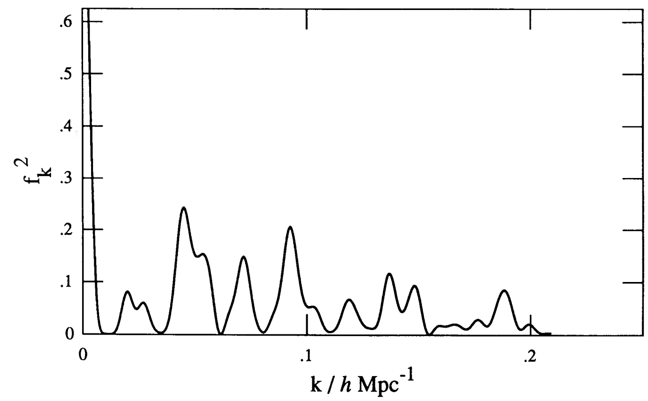

FIG. 5b

FIG. 5.—(*a*) Depth distribution and (*b*) power spectrum of the NGP data alone. The comoving distance has been calculated assuming $\Omega = 1$, $\Lambda = 0$. The counts were rebinned into cells of constant width x, and the spectrum obtained by fast Fourier transform with allowance made for the finite width of the bins (ca. $15h^{-1}$ Mpc).

Finally, in Figures 9*a*–9*d* we show the realization for the combined deep and shallow surveys. This is to be compared with Figure 2. Again, the texture in the realization matches well what is actually observed; the shallow survey fills in the broad valley in $\mathscr{I}(z)$ near $r = 0$, and this removes the corresponding sharp jagged features found there. The mean predicted power level is $\langle f_k^2 \rangle = 0.032$ (this is midway between the prediction based on the cylindrical tube and that from the Poisson clump distribution), so the observed peak is $\simeq 6.5$ times our predicted mean. In the high peak limit, the frequency of such peaks is $\simeq 0.67h^{-1}$ Mpc, so in a sample of size $\Delta k = 0.1h$ Mpc^{-1} we would expect to see such a peak $\simeq 6\%$ of the time.

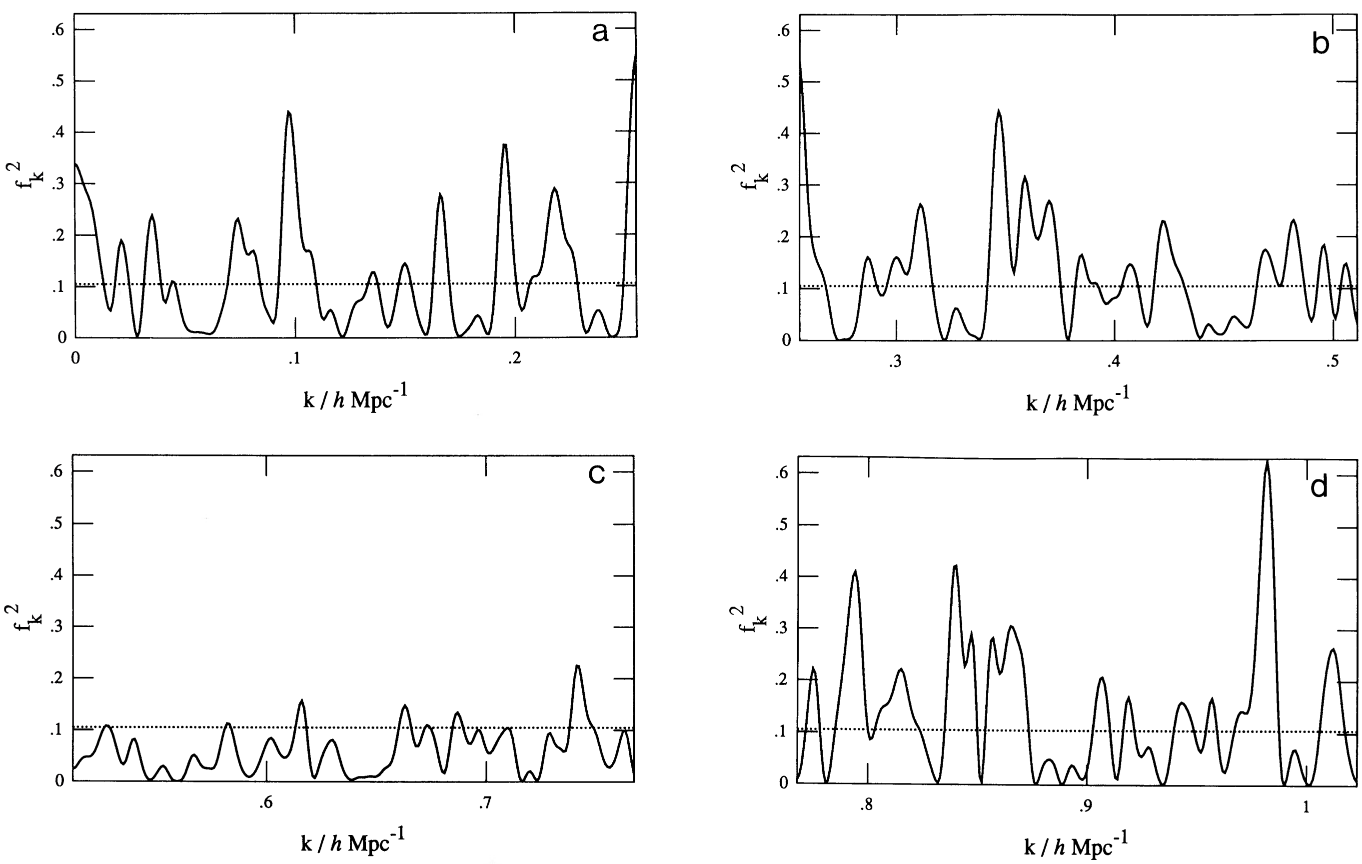

FIG. 6.—Realization of a power spectrum $P = c^2 + s^2$, where c and s are statistically homogeneous Gaussian random fields with correlation functions calculated from the survey selection functions and adopting the known power-law spectrum of small-scale galaxy clustering. We would like to stress that we are not assuming that the small-scale clustering forms a Gaussian field in *space*—that would be a terrible assumption—rather that c and s are Gaussian fields in k—an excellent approximation. The usable bandwidth for the BEKS sample is $\Delta k \simeq 0.1$, so the realization shown here is equivalent to ~ 10 independent samples of this size. It is clear that high peaks are quite common, and often appear quite isolated. Note that while the amplitude depends on the rather poorly known angular selection function, the "texture" of this random process depends only on the radial selection function ϕ, which is easily estimated from the BEKS counts. The general visual impression is very similar to the $P(k)$ shown in Fig. 5.

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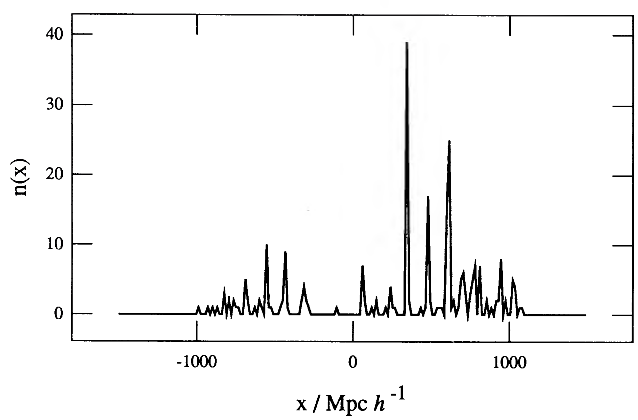

FIG. 7*a*

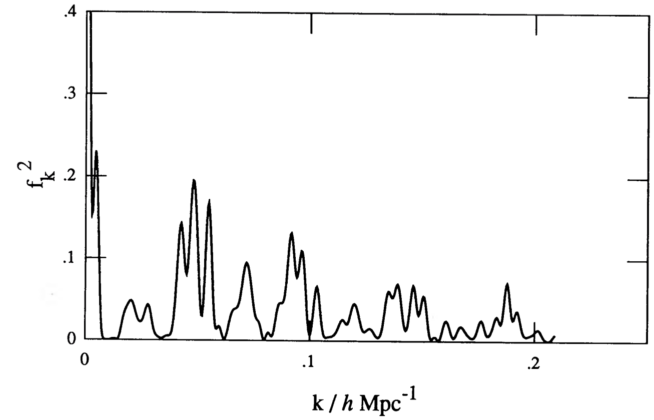

FIG. 7*b*

FIG. 7.—(*a*) Redshift distribution and (*b*) power spectrum for the NGP + SGP deep data

It is interesting that it is only with the addition of the much shallower data set that anything particularly striking appears (at least in the power-spectrum representation of the data). If the BEKS periodicity was a real feature of the universe, one would expect it to be robust with respect to the removal of $\sim 20\%$ of the survey baseline. Given that the result is in fact rather fragile, our a posteriori probability of 6% for the spike in the combined power spectrum is quite reasonable, especially as there are potentially many ways to permute data sets, and also potentially many ways in which a particular data set could be considered to be striking. In our opinion, the significance of the spike in the power spectrum is small, and would not cause us to reject the null hypothesis.

The numbers we have calculated for $\langle f_k^2 \rangle$ are not as precise as one would like; varying the number of assumed patches in the NGP to $N = 3$ and $N = 10$ changes $\langle f_k^2 \rangle$ by about 25%. A similar uncertainty is probably inherent in our assumed correlation

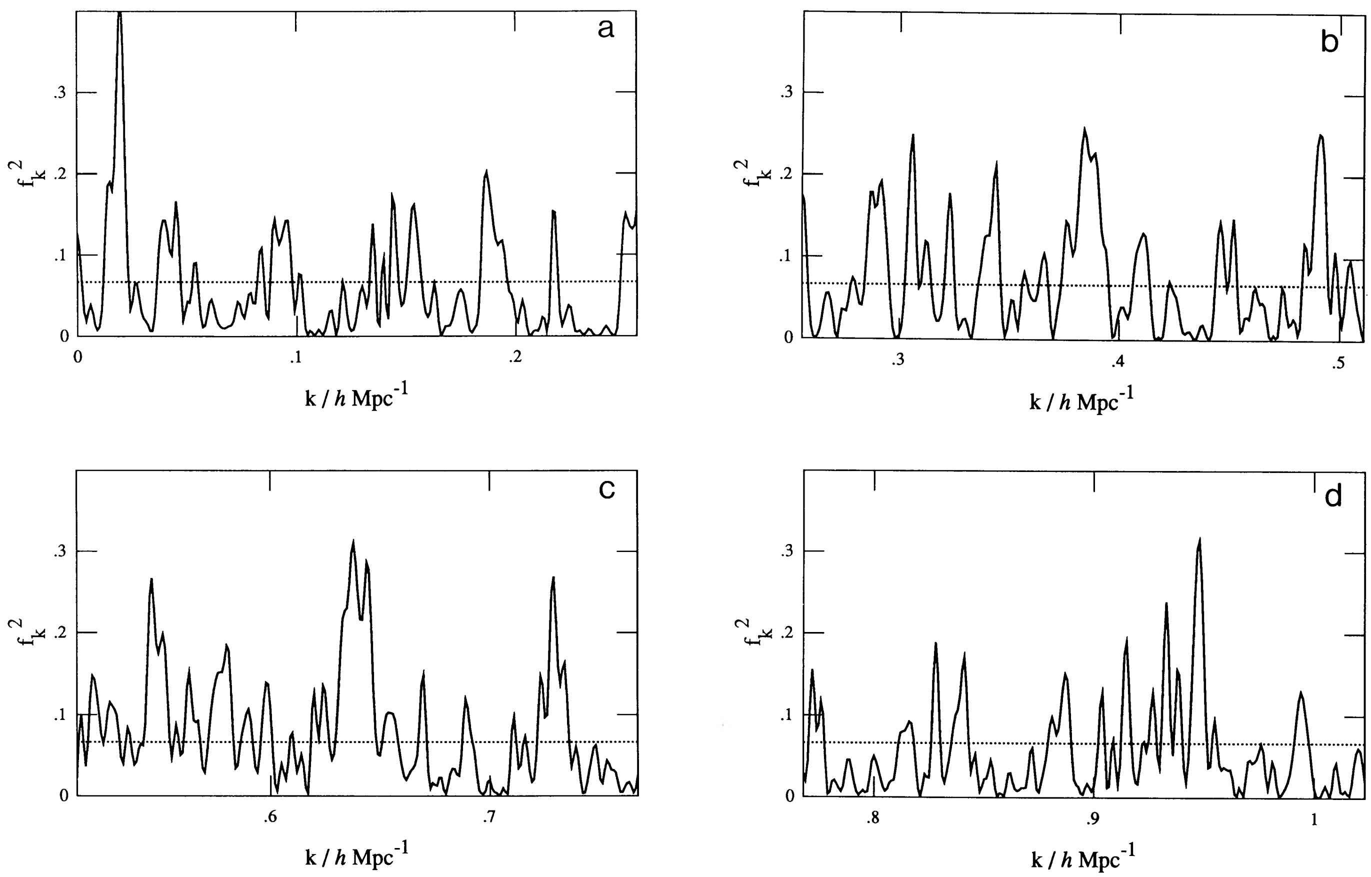

FIG. 8.—Realizations of power spectra corresponding to the SGP + NGP intensity function. Again, note the good match in appearance to the data in Fig. 7. It is interesting that while there is a visual suggestion in the counts of some periodicity, the corresponding feature in the power spectrum around $k \simeq 0.05h$ Mpc^{-1} is not at all impressive, and that in the simulation much more striking peaks abound. Insofar as the power spectrum is a reasonable way to quantify evidence for periodicity, one must conclude that in the combined BEKS deep surveys any such evidence is absent.

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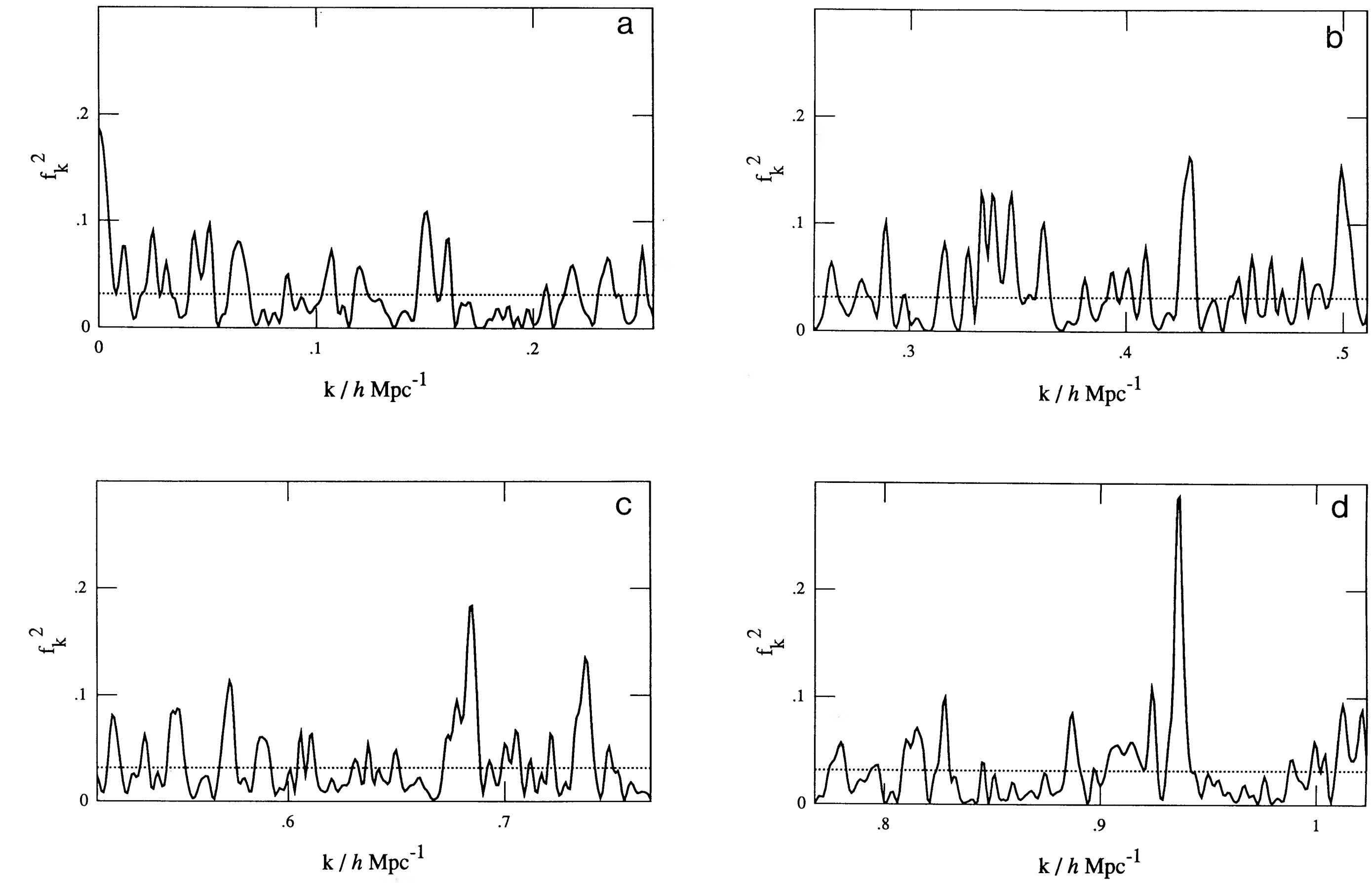

FIG. 9.—Realizations designed to match the entire BEKS sample (i.e., the deep surveys augmented by a selection of previously published surveys). This is to be compared with Fig. 2 or Fig. 2*b* of BEKS—the realization here corresponding to about 10 times the usable bandwidth there. However, note that no velocity smoothing applies in this case. As with the deep survey simulations, the texture of the field matches well what is observed. Recall that this process is simply the sum of the squares of two approximately independent Gaussian fields, with coherence properties controlled by the depth of the surveys. We were greatly impressed to see spikes appearing quite naturally from this process; our assessment of the significance of the BEKS spike was suitably modified thereafter.

length for the small-scale clustering. There is an important lesson here: one should not take very seriously any small discrepancy between predicted and observed mean power, since this probably illustrates how difficult it is to provide an accurate prediction. In fact, one should probably be pleasantly surprised at the degree of agreement between theory and the BEKS data.

4.1. *Detectability of Large-Scale Power with Pencil-Beam Surveys*

From equation (3.29), we find that in order to generate additional power equal to that aliased from small scales we need

$$\int_k dk\, kP(k) = 2\pi\langle f_k^2\rangle \left[\int dz\, z^2\phi(z)\right]^2 \Bigg/ \int dz\, z^4\phi^2(z) \simeq 283h^{-1}\ \text{Mpc}\ . \tag{4.6}$$

From equation (3.38), we find, for the selection function model above, $k_c \simeq 7 \times 10^{-3}$, so over an interval of $\Delta k = 0.1$ we should be able to measure the power to a precision (1 σ fractional error) of about 25%. Insofar as this is a small uncertainty, we can expect independent estimates of the mean power to be roughly Gaussian-distributed. Assuming (very optimistically) that the expected mean power can be predicted exactly by the procedure used here, then in the absence of any real long-wavelength power we would find fluctuations greater than about 1.5 times the mean power level about 5% of the time. Consequently, an intrinsic large-scale power level of $\simeq\langle f_k^2\rangle/2$ would, in an ideal world, be *typically* detectable at the 95% level: i.e., if the actual power added by the long-wavelength fluctuations just happens to be equal to the mean value, this would show up at the 2 σ level. However, since there are fluctuations in the actual power contributed by the long waves, one might be "lucky" and "detect" an excess of power even though typically no significant excess would appear. According to Sod's law, it is more likely that one will be unlucky. If we ask what power is needed so that 95% of the time we would find an excess at the 95% significance level, then we find we would need to add power enough to at least double $\langle f_k^2\rangle$. While this double degree of protection might seem a bit conservative, one should bear in mind that realistically the estimate of the mean power level will be rendered uncertain by such factors as the uncertainty in the appropriate galaxy correlation length and uncertainties in the selection functions, so requiring an apparent excess of power several times that estimated to arise from small-scale clustering does not seem to us to be at all overconservative.

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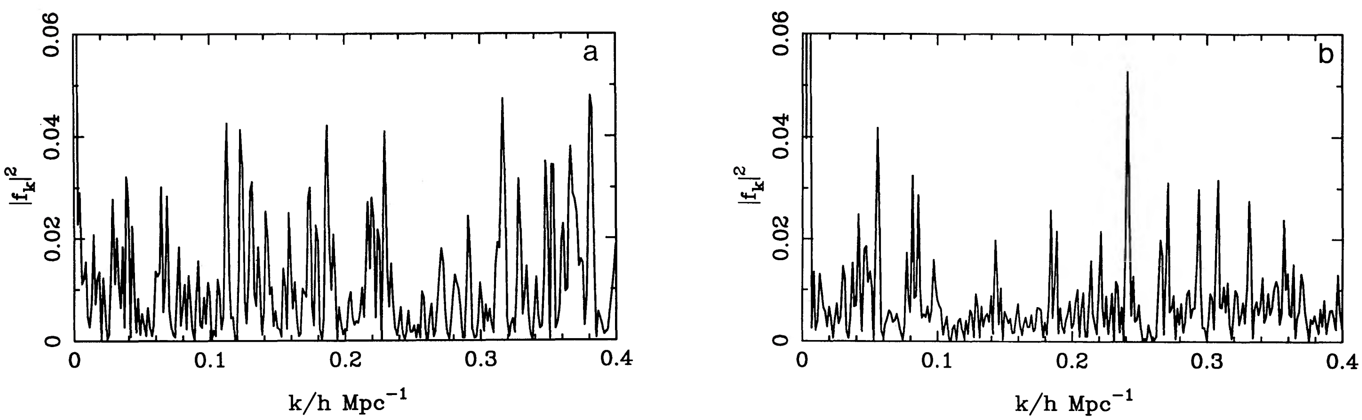

FIG. 10.—Power spectrum of (*a*) peaks of the galaxy distribution and (*b*) a clipped histogram. The peaks were obtained by passing the counts for the whole data set through a Gaussian filter of radius $R_F = 3h^{-1}$ Mpc. The clipped histogram was made using only the NGP and SGP deep surveys. Allowing for the expected decrease in the signal from the clipping process, we find a limit on the long-wavelength power which is about a factor of 3 lower than that obtained from the galaxy number-weighted power spectrum. Even with this enhanced sensitivity, the power spectrum shows no evidence for any significant large-scale structure over above that predicted from small-scale clustering.

To put the required level of intrinsic large-scale power into a more meaningful context, imagine that we insert the extra power as a "shell" in the three-dimensional spectrum at some wavenumber k_1: i.e., $P(k) = P_1\,\delta_D(k - k_1)$. Just to equal the small-scale power requires $P_1 = 283(k/h\ \mathrm{Mpc}^{-1})^{-1}$ in our units, and (modulo assumptions about how strongly biased are the galaxies) this would provide an rms density perturbation of $\langle\sigma^2\rangle^{1/2} = 0.84(\lambda/128h^{-1}\ \mathrm{Mpc})^{-1/2}$, and a three-dimensional velocity dispersion of $\langle v^2\rangle^{1/2} = 1700(\lambda/128h^{-1}\ \mathrm{Mpc})^{1/2}$ km s^{-1}. Of course, if we require, say, twice this for a significant detection, we see that the outlook for detecting large-scale density fluctuations from such surveys is rather bleak.

4.2. *Noise Suppression*

We have seen that the one-dimensional power spectrum at some wavenumber k_1 receives contributions from three-dimensional modes with $|\boldsymbol{k}| \simeq k_1$ and from modes with $|\boldsymbol{k}| \gg k_1$ which leak in through the extended sidelobes of the response functon. We have also seen how one can calculate and then subtract off the *mean* contribution from the unwanted high-frequency modes. However, the *fluctuations* in this corrected estimate of the genuine long-wavelength power at any k_1 is roughly the uncorrected power, which is very probably dominated by the leaked high-$\boldsymbol{k}$ power. Here we will explore the possibility of reducing this "noise" relative to the wanted signal by performing a nonlinear transformation on the counts in the spatial domain. The two possibilities we have examined are (*a*) smoothing the counts with a Gaussian filter and then locating peaks and (*b*) clipping the histogram of counts in redshift bins, so occupied cells are given equal weight regardless of occupation number. The resulting power spectra (details below) are shown in Figures 10*a* and 10*b*, which show that the power is reduced dramatically. Moreover, the coherence length in *k*-space is also reduced, which will render the uncertainty in any estimate of the mean power over some finite bandwidth smaller still. The burning question is: How much is the low-k_1 contribution from modes with $|\boldsymbol{k}| \simeq k_1$ changed? Only if this is reduced by less than the total power will anything have been gained. One would also like to know: Is there some transformation which will optimize the signal-to-noise ratio? The answers to these questions depend on the character of the small-scale clustering and on how low- and high-frequency modes are coupled, the latter issue being rather speculative and model-dependent.

In terms of the Poisson clump model, it is easy to see how a modest suppression of the noise might be achieved. The observed richness of a given clump is largely an accident of the exact placement of the survey beam, and could change greatly if the beam was moved by a small amount; however, the *locations* of clumps would be rather insensitive to this process, and a better approach would therefore be to weight all clumps uniformly. What we have in mind is a model whereby the probability that an infinitesimal cell in z contains n_i galaxies is the product of a clump richness distribution and the probability of finding a clump, the latter being modulated by long-wavelength perturbations Δ:

$$P(n_i)dn_i = dn_i\,\phi_i(n_i)(1 + b_i\,\Delta_i)\,. \tag{4.7}$$

Furthermore, it should be a good approximation for small Δ to treat the function b as a constant (the linear response of the system). In general, b will have contributions from clustering dynamics and also from statistics (the richest clumps being more clustered, according to the usual threshold model: see Bardeen et al. 1986, hereafter BBKS). However, as mentioned above, the true three-dimensional richness will be very poorly correlated with the numbers of galaxies seen in a skewer, so it should be a good approximation to treat all clumps as equally valid tracers of the overall light distribution, i.e., with $b \simeq 1$.

We gain in two ways if we weight clumps equally. Number weighting of clumps means the signal is dominated by the few largest clumps, which will tend to be at the peak of the selection function. Equal weighting therefore increases the number of effectively independent points (reducing $\langle f_k^2\rangle$ and increasing the sensitivity to power at a given k) and also increases the effective depth of the survey (reducing k_c and further increasing the sensitivity to power distributed across a band in k).

There are two ways in which these ideas can be implemented in practice; either we can focus on peaks in the redshift distribution, or we can process the redshift histogram in some nonlinear way to pick out clumps—the simplest method being to "clip" the

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histogram by assigning equal nonzero weight only to occupied cells. This latter method is more amenable to analysis, and is discussed in some detail in Appendx C. The presence of velocity dispersion in the clumps means that our assignment of galaxies to clumps will depend on how we filter the data. We have at our disposal a degree of freedom which corresponds to the width of the redshift bins we use, or to the width of the filter function we employ to locate peaks. In the limit that this width is very small, each galaxy is a clump and we recover the standard method. If the width is chosen too large, however, we lose power, as there will be few peaks and few unoccupied cells in the clipped histogram. The correct filter size must be approximately matched to small-scale clustering; a few megaparsecs seems appropriate. In practice, a factor of 2 variation in this scale seems to make little difference to the results.

We show some results for clump power spectra in Figure 10. This compares (*a*) f_k^2 for peaks found using a Gaussian filter of standard deviation $3h^{-1}$ Mpc on the synthesized total data set of Figure 2, with (*b*) f_k^2 for the clipped version of the NGP + SGP histogram (using the BEKS bins of 0.005 in redshift). In both cases, the mean power is very low: $\langle f_k^2 \rangle \lesssim 0.01$, and neither plot displays any evidence for extra power near $k = 0.05$. For both cases, the coherence length k_c should be close to the Nyquist value for the $z = \pm 0.5$ box: $k_c = (2\pi/2202)h$ Mpc^{-1} (cf. Appendix C). Thus, the rms noise now corresponds to a value of Δ_{1D}^2 of about 0.2 at $k = 0.05$; at $k < 0.1$ there are about 35 independent modes, so we would expect a 1 σ sensitivity of about 17% of this mean power in averaging over a band. According to our previous criterion of a "safe" 95% detection, therefore, $\Delta_{1D}^2 \simeq 0.14$ would have been detectable—a somewhat more interesting number than our previous $\Delta_{1D}^2 \simeq 0.84$.

The increase in sensitivity obtained by down-weighting the richer clumps or peaks is quite substantial. It is however, conditional on the validity of the model (4.7), in which clumps are drawn from some universal clump-mass distribution and placed at positions which sample the long-wave density field Δ. If this is correct, it should come as no surprise that unequally weighting clumps (i.e., by mass) will decrease the effective number of clumps and correspondingly decrease the sensitivity of the survey. As we have seen, however, the improvement in sensitivity is limited to a factor ~ 2 in clustering *amplitude*, so there is no reason to modify our general conclusion concerning the efficiency of one-dimensional surveys in constraining large-scale structure.

4.3. *Coherent Peculiar Velocities and One-dimensional Power Spectra*

In the foregoing calculations, we have implicitly assumed that coherent velocities associated with large-scale clustering have a negligible effect, and that the structure in redshift space is essentially identical to that in real space. This should be a good approximation if the universe has a low density, $\Omega \ll 1$, or if the galaxies are strongly biased. In that case, we are seeing a one-dimensional slice of a three-dimensional homogeneous random process, and the one-dimensional power spectrum is a very simple projection (see eq. [3.2]). Thus, even if there are sharp features in the three-dimensional power spectrum, they will tend to be washed out. In a high-density universe (with unbiased galaxies), however, peculiar velocities can have a large distortion effect on the clustering pattern, and this needs to be taken into account in interpreting one-dimensional power spectra. Kaiser (1987) has discussed the implications of peculiar velocity distortion mainly with application to the correlation function. Here we give some analogous results which are more directly relevant to power spectra.

For a statistically isotropic system, the power spectrum of a one-dimensional sample is simply related to the three-dimensional power by

$$P_{1D}(k_z) = \frac{1}{(2\pi)^3} \int d^2k_\perp P_{3D}(k_z^2 + k_\perp^2) = \frac{1}{2\pi} \int_{k_z}^{\infty} dk\, k P_{3D}(k) . \tag{4.8}$$

It follows from this that if $P_{3D}(k)$ is a delta function at k_1, then P_{1D} will be flat for $0 < k < k_1$ and zero otherwise, and that the one-dimensional power spectrum is a monotonically decreasing function of k. Thus, for a statistically isotropic system, the expected power spectrum can never have a peak [though this need not be true for an estimate of $P(k)$ obtained from a finite data set] and generally one would expect the one-dimensional power spectrum to be rather featureless. In redshift space, however, things are rather different; the power spectrum is now anisotropic and we have (in three dimensions)

$$P_z(k) = P_r(k)(1 + \alpha\mu^2)^2 , \tag{4.9}$$

where z and r denote redshift and real space respectively, $\alpha = \Omega^{3/5}/b$ (b being the bias parameter), and μ is the cosine of the angle between the line of sight and the wavevector. The projection of this anisotropic power spectrum into one dimension is now

$$P_{1D}(k_z) = \frac{1}{(2\pi)^2} \int d^2k_\perp P_{3D}(k_z^2 + k_\perp^2)\left(1 + \alpha \frac{k_z^2}{k_z^2 k_\perp^2}\right)^2 = \frac{1}{2\pi} \int_{k_z}^{\infty} dk\, k P_{3D}(k)\left(1 + \alpha \frac{k_z^2}{k^2}\right)^2 . \tag{4.10}$$

If we consider again the extreme example of a delta-function shell of power in three dimensions, we find that

$$P_{1D}(k) \propto (1 + \alpha k^2/k_1^2)^2 \qquad \text{for} \quad k < k_1. \tag{4.11}$$

This is the same as the $\alpha = 0$ result for $k \to 0$, but for $\alpha = 1$, for example (an unbiased, $\Omega = 1$ model), the one-dimensional power spectrum rises to a sharp peak at $k = k_1$, which is 4 times the power predicted for $\alpha = 0$. This rather extreme example shows that it is possible to get nonmonotonic one-dimensional power spectra, and, in principle at least, one can have quite strong features in the expected one-dimensional power spectrum [though we caution that given the naturally spiky nature of power spectra, it may be difficult to obtain sufficient data to confirm the existence of a real feature in $P_{1D}(k)$ even if it is there]. This process will also increase the power averaged over some wave band; the integrated power is increased by a factor 28/15 for $\alpha = 1$, but for $\alpha = 0.5$ (e.g., $\Omega = 1$ and $b = 2$), which is probably more reasonable for these galaxies, the increase in integrated power is only about 40%.

We should stress that the foregoing analysis applies only to intrinsically long-wavelength fluctuations, and not to high-frequency power which is aliased to low k; for such modes one need only worry about the smearing effect of random peculiar velocities. For

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the BEKS data the foregoing discussion is of academic interest only: the low-k power here is almost certainly dominated by small-scale clustering. However, the available data will certainly increase with time, and eventually such effects may need to be considered.

5. DISCUSSION

In the above sections we have discussed the power spectrum of the BEKS one-dimensional redshift survey. Our principal conclusion is that both the amplitude and the relative isolation of the peak at $\lambda = 128h^{-1}$ Mpc in their power spectrum are well consistent with a null hypothesis in which the universe is uniform apart from the known clustering on scales $\sim$1–10 Mpc. This result may be thought of as arising either via the Poissonian distribution of a relatively small number of rich clumps or equivalently as a projection of power from three-dimensional short-wavelength modes nearly perpendicular to the skewer to long wavelengths along the skewer, plus in either case the convolving effect of virialized peculiar velocities. Our conclusions differ from those of BEKS. We would argue that the reason for this is that their approximate estimate of the fluctuations in the pair counts underestimated the strength of the fluctuations for a realistic survey geometry, and that their estimate of the mean power level gave a coincidentally similar underestimate, largely as a consequence of using spatial frequencies where the power is suppressed by the finite width of the clumps. We were initially impressed by the sharpness of the spike in $P(k)$. However, our realization of power spectra (either from the Poisson clumps model of § 2 or by invoking the central-limit theorem in § 4) reveal that sharp, fairly isolated peaks are not at all uncommon, even under the rather conservative assumption that the Fourier amplitudes are Gaussian-distributed.

We have calculated the level of real extra large-scale power which might be detectable in principle using a one-dimensional redshift survey like that of BEKS. The result is so large (variance in δ of order unity of 100 Mpc scales) that it is probably already excluded by shallower surveys. Realistically, we are interested in testing for much smaller departures from an $n \simeq -1$ spectrum at large scales. For example, recall that Δ^2_{3D} has the value 0.072, at $\lambda = 128h^{-1}$ Mpc for pure $\xi \propto r^{-1.8}$ clustering; for CDM, this becomes about 0.01 (depending on h). Clearly, there is no hope of detecting this difference between two small signals with a lower dimensional survey. Establishing departures from an $n \simeq -1$ spectrum continues to be a difficult task even looking at smaller wavelengths and with larger, more powerful three-dimensional surveys (such as the QDOT *IRAS* redshift survey; Efstathiou et al. 1990).

Finally, we should point out what our calculations have *not* established. Although we have compared the BEKS results with the minimal standard model of known isotropic small-scale clustering and found no disagreement, this does not prove that unconventional pictures for large-scale structures are ruled out. One of our main results has been to show how poorly single skewer surveys can discriminate between competing models. It is therefore hardly surprising that papers have appeared claiming consistency between the BEKS data and three-dimensional Voronoi tessellation models (Coles 1990; van de Weygaert 1991). This consistency should not be taken as arguing for or against the Voronoi model. What needs to be done now is to expand the dimensionality of the data by surveying adjacent skewers. The interpretation of the resulting data may not be straightforward, however, since correlated spikes in adjacent skewers are certainly expected in conventional models for large-scale structure. Even hierarchical models like CDM display large-scale structure dominated by "walls" and "filaments" on scales of tens of megaparsecs (Park 1990); these structures are largely responsible for the observed small-scale clustering. The challenge will be to see what degree of coherence such features display on larger scales.

These are possibilities for the future. For the present, we emphasize that the BEKS data seem to present no difficulty for conventional models where the large-scale density is approximately a random Gaussian field with little power on scales $\sim$100 Mpc.

J. A. P. thanks CITA for their hospitality while this work was carried out and acknowledges additional financial support from CIAR. N. K. acknowledges helpful discussions with Alex Szalay, and support from NSERC and the CIAR.

APPENDIX A

REALISTIC SURVEY GEOMETRY

This observer takes a one-dimensional Fourier transform of the counts

$$f_k(\boldsymbol{r}_0) = \frac{1}{N_{\rm gals}} \int dz\, n(z) \exp{(ikz)} \ . \tag{A1}$$

Now, neglecting shot noise from galaxies,

$$n(z) = \alpha \iint dx\, dy\, \phi(z) W(x/z,\, y/z)[1 + \delta(\boldsymbol{r}_0 + \boldsymbol{r})] \ , \tag{A2}$$

where

$$\alpha = \left[d\Omega \int dz\, z^2 \phi(z)\right]^{-1} , \qquad d\Omega = \int d^2\theta\, W(\theta) \ , \tag{A3}$$

so

$$f_k(\boldsymbol{r}_0) = \int d^3r\, g(\boldsymbol{r})[1 + \delta(\boldsymbol{r}_0 + \boldsymbol{r})] \ , \tag{A4}$$

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i.e., f_k is the convolution of $1 + \delta(\boldsymbol{r})$ with the window function

$$g(\boldsymbol{r}) = \alpha\phi(z) \exp (ikz)W(x/z, y/z) . \tag{A5}$$

Consequently,

$$f_k(\boldsymbol{r}_0) = \frac{1}{(2\pi)^3} \int d^3k\, \delta(\boldsymbol{k})\tilde{g}(-\boldsymbol{k}) \exp (i\boldsymbol{k} \cdot \boldsymbol{r}_0) , \tag{A6}$$

where the response function is

$$\tilde{g}(\boldsymbol{k}) = \int d^3r\, g(\boldsymbol{r}) \exp (i\boldsymbol{k} \cdot \boldsymbol{r}) = \alpha \int dz\, z^2\phi(z) \exp [i(k + k_z)z]\tilde{W}(zk_x, zk_y) , \tag{A7}$$

and

$$\tilde{W}(\kappa_x, \kappa_y) \equiv \iint d\theta_x d\theta_y\, W(\theta_x, \theta_y) \exp [i(\kappa_x \theta_x + \kappa_y \theta_y)] . \tag{A8}$$

The general features of the response section are the same as those of the idealized experiment discussed in § 3.1. To calculate the covariance matrix elements, we write

$$c(k) = \frac{\alpha}{2\pi} \int dk_z \int dz\, z^2\phi(z) \cos (kz) \exp (-ik_z z)\Delta(k_z, z) , \tag{A9}$$

where now

$$\Delta(k_z, z) = \frac{1}{(2\pi)^2} \iint dk_x dk_y\, \tilde{W}(zk_x, zk_y)\delta(k_x, k_y, k_z) \exp (i\boldsymbol{k} \cdot \boldsymbol{r}_0) . \tag{A10}$$

The covariance function for $c(k)$ is now

$$\langle c(k)c(k')\rangle = \frac{\alpha^3}{(2\pi)^2} \int dk_z \int dz \int dk'_z \int dz' z^2\phi(z)z'^2\phi(z') \cos (kz) \cos (k'z') \exp [-i(k_z z - k'_z z')]\langle\Delta(k_z, z)\Delta(-k'_z, z')\rangle , \tag{A11}$$

but

$$\begin{aligned}\langle\Delta(k_z, z)\Delta(-k'_z, z')\rangle &= \frac{1}{(2\pi)^4} \iint d^2k\, d^2k'\, \tilde{W}(zk_x, zk_y)\tilde{W}^*(z'k'_x, z'k'_y) \exp [i(\boldsymbol{k} - \boldsymbol{k}') \cdot \boldsymbol{r}_0]\langle\delta(\boldsymbol{k})\delta^*(\boldsymbol{k}')\rangle \\ &= \frac{\delta(k_z - k'_z)}{(2\pi)^2} \int d^2k\, \tilde{W}(zk_x, zk_y)\tilde{W}^*(z'k_x, z'k_y)P(k_x, k_y, k_z) ,\end{aligned} \tag{A12}$$

so

$$\langle c(k)c(k')\rangle = \frac{\alpha^2}{(2\pi)^3} \int dz \int dz' z^2\phi(z)z'^2\phi(z') \cos (kz) \cos (k'z') \int d^2k\, \tilde{W}(zk_x, zk_y)\tilde{W}(z'k_x, z'k_y) \int dk_z\, P(k_x, k_y, k_z) \exp [i(z - z')k_z] . \tag{A13}$$

We are primarily interested in leakage of power for modes with $k \sim H/\theta$. For these, $P(k)$ is effectively independent of k_z, so, taking $P(k)$ outside of the last integral, we have

$$\langle c(k)c(k')\rangle = \frac{\alpha^2}{2\pi} \int dz\, z^4\phi^2(z) \cos (kz) \cos (k'z) \int dk\, k\tilde{W}^2(zk)P(k) . \tag{A14}$$

For $P(k) = P_0 k^n$ the integrals separate, and for $k, k' \gg 1/L$, we obtain

$$\langle c(k)c(k')\rangle = \langle s(k)s(k')\rangle = \frac{\langle f^2\rangle}{2} \psi_+(k - k') = \frac{\langle f^2\rangle}{2} \int dz\, w^2(z) \cos [(k - k')z] \tag{A15}$$

$$\langle c(k)s(k')\rangle = \frac{\langle f^2\rangle}{2} \psi_-(k - k') = \frac{\langle f^2\rangle}{2} \int dz\, w^2(z) \sin [(k - k')z] , \tag{A16}$$

where

$$\langle f^2\rangle \equiv \frac{P_0}{2\pi} \frac{\int dz\, z^{2-n}\phi^2(z)}{[\int dz\, z^2\phi(z)]^2} \frac{\int d\kappa\, \kappa^{1+n}\tilde{W}^2(\kappa)}{[\int d^2\theta\, W(\theta)]^2} , \tag{A17}$$

and

$$w(z) \equiv z^{1-n/2}\phi(z)\bigg/\left[\int dz\, z^{2-n}\phi^2(z)\right]^{1/2} . \tag{A18}$$

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We will also be interested in the power arising from long-wavelength fluctuations. For these we have $kz \ll H/\theta$, so we can set $\tilde{W}(kx) = \tilde{W}(0)$ in equation (A14), to obtain

$$\langle c(k)c(k')\rangle = \frac{1}{(2\pi)^2} \frac{\int dz\, z^4 \phi^2(z) \cos(kz) \cos(k'z)}{[\int dz\, z^2 \phi(z)]^2} \int dk_x \int dk_y P(k_x, k_y, k) , \tag{A19}$$

so for the expectation for the power we have

$$\langle f_k^2 \rangle = \frac{1}{2\pi} \frac{\int dz\, z^4 \phi^2(z)}{[\int dz\, z^2 \phi(z)]^2} \int_k dk\, kP(k) , \tag{A20}$$

where the integration over $P(k)$ can be taken only over modes with $k \ll H/\theta$. For higher frequency modes we must use equation (A17). This neglects the effects of coherent peculiar velocities which may be important if Ω is large and the galaxies are not strongly biased. This is discussed in § 4.3.

APPENDIX B

PEAKS IN POWER SPECTRA

In the situation of interest, we have the Fourier transform f_k of some statistically homogeneous process $g(r)$, which we view through some "window" $w(r)$. The real and imaginary parts of f_k are then

$$c(k) = \int w(x)g(x) \cos(kx) dx , \qquad s(k) = \int w(x)g(x) \sin(kx) dx . \tag{B1}$$

In the context of the Poisson clump model, for instance, we assumed that there were clusters drawn from some distribution of intrinsic richness, and that these clusters were uniformly distributed in x. The observed richness of each clump was given by the intrinsic richness multipled by a smoothed galaxy density distribution dp/dx. In this model we would identify $w(x)$ with dp/dx and the homogeneous stochastic process $g(x)$ with a homogeneous Poissonian point process with the strengths of the "shots" drawn from the richness distribution. In § 3 we showed that assuming a power-law power spectrum for galaxy clustering, and a conical survey, the two-point correlation functions for $c(k)$ and $s(k)$ were identical to those of an arbitrary homogeneous stochastic process $g(x)$ multiplied by a window function $w(x) \propto x^{1-n/2}\phi(x)$. Note that the windows in these cases are quite similar [but not identical; the smoothed dp/dx being proportional to $x^2\phi(x)$]. Similarly, in the model of § 3.1, we would identify $g(x)$ with the continuous δ field, and $w(x)$ with the longitudinal profile of the window. Now provided there are enough independent fluctuations of the stochastic field g within the window, we can apply the central-limit theorem to show that c and s will become Gaussian-distributed, and hence the statistical properties depend only on two-point correlation functions $\langle c(k)c(k')\rangle$, etc. Moreover, we showed in §§ 3.1 and 3.2 that for the wavenumbers of interest these correlation functions depend only on $\Delta k \equiv k - k'$, so the $c(k)$ and $s(k)$ fields themselves become a pair of statistically homogeneous Gaussian random fields. In this Appendix, we will calculate the frequency of peaks of the random field $R^2(k) = c(k)^2 + s(k)^2$.

The frequency of peaks of a *single* Gaussian random field [the primordial $\delta(\boldsymbol{r})$ field] has been widely considered in a cosmological context. There the correlation information enters only through moments of the power spectrum $P(k)$. Here we have an analogous situation, only now the statistics of the peaks of R are determined by a set of moments over what we have termed (in § 3.2) the *intensity* defined as follows (cf. BBKS):

$$\sigma_m^2 \equiv \frac{1}{2} \int \mathscr{I}(x) x^{2m} dx , \tag{B2}$$

where $\mathscr{I}(x) \propto w^2(x)$. These moments will be identical for both c and s fields; σ_0 is the rms of the fields.

The problem we are considering here was tackled by Rice (1954). The problem is complicated by the fact that c and s are not in general independent: their derivatives are correlated as follows (the prime denoting k-space differentiation):

$$\langle cs' \rangle = \sigma_{1/2}^2 , \qquad \langle c''s' \rangle = -\sigma_{3/2}^2, \text{ etc.} \tag{B3}$$

The fields can only be treated as independent for a symmetric selection function, which is not generally the case. Rice's treatment concentrates on the symmetric case and is also rather involved; we shall give a simpler approach to the problem.

In looking for peaks in $R \equiv c^2 + s^2$, we can exploit the fact that the problem contains two "gauge" degrees of freedom: transformations which do not affect the power $R^2(k)$. One of these is the ability to define new random fields via a rotation in (c, s)-space:

$$\begin{pmatrix} c \\ s \end{pmatrix} \to \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} . \tag{B4}$$

Such a transformation preserves correlations between c, s and their derivatives; we can therefore without loss of generality consider peaks in R to occur at $s = 0$. Another way of seeing this is to note that the frequency of peaks in some infinitesimal rectangle $dc\, ds$ around (c, s) depends by symmetry only on R^2; having calculated this for the element around $c = R$, $s = 0$, we simply multiply the result by $2\pi R/ds$ to obtain the frequency of peaks in the annulus R: $R + dR$. The second degree of freedom is translation of the

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spatial origin: As discussed in the main text, this also induces a rotation in the (c, s)-plane, but this is k-dependent, causing f_k to spiral around the k-axis, introducing correlations between even (odd) derivatives of c and odd (even) derivatives of s. We choose to place our spatial origin at the center of mass of $\mathscr{I}(x)$, so that $\sigma_{1/2}^2 = 0$. This decouples c and s' and we only need to deal with the correlations between c'' and s'.

Finding peaks in R requires the distribution of R plus first and second derivatives (with respect to k-space); with $s = 0$, these depend only on (c, c', c'', s'), all of which are independent of s. So the number density of stationary points in R in the range $dc\, ds$ is given by a simple modification of the usual expression:

$$\frac{\partial^2 N}{\partial c\, \partial s} = \int_{-\infty}^{\infty} \int p(c = R,\, c' = 0,\, c'',\, s = 0,\, s') \left| \frac{\partial c'}{\partial R'} \right| |R''|\, dc''\, ds'$$

$$= \int_{-\infty}^{\infty} \int p(c = R,\, c' = 0,\, c'',\, s = 0,\, s') \left| c'' + \frac{s'^2}{c} \right| dc''\, ds' \,. \tag{B5}$$

For peaks in R, the regime of integration is $R'' < 0 \Rightarrow c'' < -s'^2/c$. The required probability density is a multivariate Gaussian, where s and c' are independent variables, with variances σ_0^2 and σ_1^2, respectively, and the correlation matrix for (c, c'', s') is

$$M = \begin{pmatrix} \sigma_0^2 & -\sigma_1^2 & 0 \\ -\sigma_1^2 & \sigma_2^2 & -\sigma_{3/2}^2 \\ 0 & -\sigma_{3/2}^2 & \sigma_1^2 \end{pmatrix} . \tag{B6}$$

We can tidy all this up, first by using symmetry to integrate round a ring in (c, s)-space to convert $\partial^2 N/\partial c\, \partial s$ to $(2\pi R)^{-1} \partial N/\partial R$, and second by introducing BBKS–like variables:

$$\nu \equiv R/\sigma_0 \,, \qquad x \equiv -c''/\sigma_2 \,, \qquad y \equiv s'/\sigma_1 \,, \qquad \gamma \equiv \sigma_1^2/(\sigma_0 \sigma_2) \,, \qquad \epsilon \equiv \sigma_{3/2}^2/(\sigma_1 \sigma_2) \,, \qquad k_* \equiv \sigma_1/\sigma_2 \,. \tag{B7}$$

We now obtain

$$\frac{\partial N}{\partial \nu} = (2\pi)^{-3/2} \frac{\nu}{\sqrt{1 - \gamma^2 - \epsilon^2}} k_*^{-1} \exp\left(-\frac{\nu^2}{2}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{y^2}{2}\right) dy \int_{\gamma y^2/\nu}^{\infty} \left(x - \frac{\gamma y^2}{\nu}\right) \exp\left[-\frac{1}{2} \frac{(x - \gamma\nu - \epsilon y)^2}{1 - \gamma^2 - \epsilon^2}\right] dx \,. \tag{B8}$$

Performing the x integration leaves us with

$$\frac{\partial N}{\partial \nu} = \frac{\nu}{(2\pi)^{3/2} k_*} \exp\left(-\frac{\nu^2}{2}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{y^2}{2}\right) dy \left\{ a \exp\left(-\frac{b^2}{2a^2}\right) + b \left(\frac{\pi}{2}\right)^{1/2} \left[1 + \mathrm{erf}\left(\frac{b}{\sqrt{2}a}\right)\right] \right\} , \tag{B9}$$

where $a \equiv (1 - \gamma^2 - \epsilon^2)^{1/2}$ and $b \equiv \gamma\nu + \epsilon y - \gamma y^2/\nu$. To obtain $N(>\nu)$, it therefore appears necessary to perform two numerical integrations, making this a rather messier problem than a single one-dimensional field where the integral peak density can be obtained analytically (Cartwright & Longuet-Higgins 1956). Nevertheless, some useful approximations can be obtained.

For $\gamma\nu \gg 1$, the term in square brackets goes to $(2\pi)^{1/2}\gamma\nu$, and we obtain the integral density for high peaks

$$N(>\nu) = \frac{\gamma}{(2\pi)^{1/2} k_*} \nu \exp\left(-\frac{\nu^2}{2}\right) , \tag{B10}$$

independent of ϵ. For the uncorrelated case, this is in agreement with the density of upcrossing points according to Coles & Barrow (1987). The differential density can be found exactly in the cases $\gamma = 0$ or 1, where again it is independent of ϵ:

$$\frac{\partial N}{\partial \nu} = \frac{1}{2\pi k_*} \nu \exp\left(\frac{-\nu^2}{2}\right) \qquad (\gamma = 0)$$

$$= \frac{1}{2\pi k_*} \left[2\nu \exp(-\nu^2) + (2\pi)^{1/2}(\nu^2 - 1) \exp\left(-\frac{\nu^2}{2}\right) \mathrm{erf}\left(\frac{\nu}{\sqrt{2}}\right)\right] \qquad (\gamma = 1) \,, \tag{B11}$$

implying total densities of $(2\pi k_*)^{-1}$ and $(\pi k_*)^{-1}$, respectively. The total density is not generally obtainable, but an excellent approximation (to $\sim$0.1%) is

$$N = \frac{1 + \gamma^{1.44}(1 - |\epsilon|)^{0.2(1-\gamma)^{1.5}}}{2\pi k_*} . \tag{B12}$$

Some plots of $f(>\nu)$ based on this equation plus the high peak integral density are compared with the exact probability in Figure 11. These have been converted into power terms, where the expectation of the power is $\langle P \rangle = \langle R^2 \rangle = 2\sigma_0^2$, so that

$$\frac{P}{\langle P \rangle} = \frac{\nu^2}{2} . \tag{B13}$$

Interestingly, the net result of these calculations is that the effect of asymmetry in the selection function is always small. Even for the maximum ϵ allowed by the constraint that det $M = (1 - \gamma^2 - \epsilon^2) > 0$, the perturbation to $N(>\nu)$ is only a few percent. For

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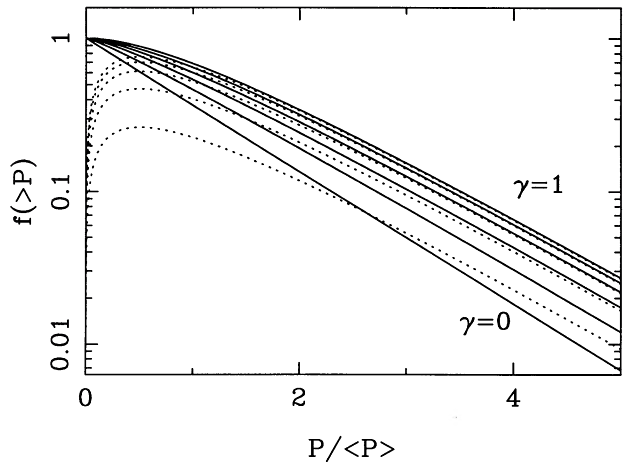

FIG. 11.—Integral probability distributions for the height of peaks in $P = c^2 + s^2$, where c and s are random Gaussian fields. Curves are shown for $\gamma = 0, 0.2, 0.4, 0.6, 0.8, 1$, where γ describes the degree of concentration of the intensity function for the c and s fields. The dotted line shows the high-peak limit, which is generally an excellent approximation provided that $\gamma \gtrsim 0.3$. Allowing the asymmetry parameter ϵ to be nonzero produces an almost indiscernible alteration to this figure, changing $f(>P)$ by only a few percent around $P \simeq \langle P \rangle$.

selection functions which have less pathological deviations from symmetry, treating c and s as independent is a very good approximation indeed. Finally, note that the high-peak limit seems to work very well for $P/\langle P \rangle \gtrsim 1$ (with the exception of very low γ), and moreover is conservative in the sense of providing a lower bound to $f(>P)$.

APPENDIX C

CLIPPED-HISTOGRAM POWER-SPECTRUM ESTIMATORS

In this Appendix we consider general estimators for the power spectrum in the form of a modification of the counts in cells:

$$f_k = \sum_i F_i(n_i) \exp(ikx_i) , \tag{C1}$$

where the functions F_i are arbitrary at this stage. We will show below that, with a not unreasonable model for galaxy clustering, one can achieve a significantly more sensitive estimate of the long-wave power spectrum. In the foregoing analysis, we have implicitly assumed that the galaxy fluctuations can be split into two statistically independent components: a "noise" term from the strong small-scale clustering, and the long-wavelength "signal" that we wish to measure or constrain. For the linear estimator (2.1) we were able to express the mean "noise" power in terms of the two-point function of galaxy clustering. With the nonlinear estimator (C1), more information is needed.

One way to proceed is to say that the small-scale clustering results in counts in cells which are, to a first approximation, statistically independent, and which sample some cell "richness" distribution $\phi(n)$. This is in turn modulated by the long-wave density perturbations: i.e., the probability that the ith cell contains n_i galaxies is

$$P(n_i)dn_i = dn_i\, \phi_i(n_i)[1 + b_i(n_i)\Delta_i] . \tag{C2}$$

Here the subscript i on ϕ allows for variation of the apparent richness distribution with depth, and the function $b_i(n)$ (also allowed to be a function of position) describes how strongly the abundance of cells with a given count n are modulated by the density perturbation Δ_i. The function b will dictate the form of the function F which optimizes the power spectrum estimator. A very simple model (but possibly quite a realistic one) is to assume $b \simeq 1$. This would apply if the small-scale clustering results in a "graininess" of the galaxy distribution when viewed on a scale equal to the width of the survey cone (the function ϕ being the distribution function for grain masses), and if the long-wave perturbations displace grains without creating or destroying them. Provided that the cells are small enough that multiple occupancy can be neglected, we have a one-to-one relation between grains and occupied cells, and this model should apply. If the mean occupied cell fraction increases (e.g., owing to a larger cell length), the value for b (still assuming conserved grains) will decrease; increasing the number of grains in some region will clearly result in a smaller increase in occupied cells, since some of the additional grains will fall in already occupied cells.

If this constant-b model is appropriate, it is easy to see that weighting galaxies equally is inefficient, and that one should weight each occupied cell equally, regardless of occupation number: $F =$ constant. According to this model, grains of any mass are equally good tracers of the large-scale density field; their mass is as irrelevant to their information concerning large-scale structure as would be their color. This "clipped histogram" power spectrum gives higher precision for two reasons: first, the effective number of "objects" increases (there are 75 occupied cells in all in the BEKS deep survey histogram), and, second, the effective depth of the survey increases, giving more independent estimates of the power for any interval of wavenumber.

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This constant-F weighting scheme is only the optimum if b = constant. In the case of a broader beam survey (or longer cells) such that essentially all cells are occupied, one will clearly get a very poor estimator from a clipped histogram. In such a case it would be desirable to modify the weighting scheme to give lower weight to the low occupation number cells with lower b. We now construct a one-dimensional power-spectrum estimator for arbitrary F, and calculate the statistical uncertainty. We will then apply this to the clipped histogram case assuming b = constant, and estimate the appropriate value for b. We will then discuss how to optimize the estimator in the more general situation.

If we have the Fourier transform f_k defined above, then the average power spectrum is

$$\langle f_k^2 \rangle = \sum_i \sum_j \langle F_i(n_i) F_j(n_j) \rangle \cos\,[k(x_i - x_j)] \,. \tag{C3}$$

Now

$$\begin{aligned} \langle F_i(n_i) F_j(n_j) \rangle &= \int dn\, \phi_i(n) F_i^2(n) \langle [1 + b_i(n)\Delta_i]^2 \rangle && \text{if } j = i \,, \\ &= \int dn\, \phi_i(n) F_i(n) \int dn' \phi_j(n') F_j(n') \langle [1 + b_i(n)\Delta_i][1 + b_j(n')\Delta_j] \rangle && \text{if } j \neq i \,. \end{aligned} \tag{C4}$$

Defining

$$\begin{aligned} \Phi_i &\equiv \int dn\, \phi_i(n) F_i(n) \,, \qquad & \Phi_i^{(2)} &\equiv \int dn\, \phi_i(n) F_i^2(n) \,, \\ \Psi_i &\equiv \int dn\, \phi_i(n) b_i(n) F_i(n) \,, \qquad & \Psi_i^{(2)} &\equiv \int dn\, \phi_i(n) b_i^2(n) F_i^2(n) \,, \end{aligned} \tag{C5}$$

we find

$$\langle f_k^2 \rangle = \sum_i \left[(\Phi_i^{(2)} - \Phi_i^2) + \langle \Delta^2 \rangle (\Psi_i^{(2)} - \Psi_i^2) \right] + \sum_i \sum_j (\Phi_i \Phi_j + \langle \Delta_i \Delta_j \rangle \Psi_i \Psi_j) \cos\,[k(x_i - x_j)] \,. \tag{C6}$$

The first term in the double sum is just $\tilde{\Phi}(k)\tilde{\Phi}^*(k)$; provided that $F_i(n)$ is [like $\phi_i(n)$] a smoothly varying function of position x_i, this will only contribute for very low wavenumbers $k \sim 1/L$ and can safely be ignored. The correlation function $\langle \Delta_i \Delta_j \rangle = \xi(x_i - x_j)$ will have a width $\simeq 1/k$. For $k \gg 1/L$ we can set $\Psi_j \simeq \Psi_i$ in the second term in the double sum, so this becomes $\sum_i (\Psi_i^2/\delta l) \sum_i \delta l \langle \Delta_i \Delta_j \rangle \cos\,[k(x_i - x_j)]$ (we assume that the cell length δl is also a slowly varying function of position in order to insert this factor inside the sum on i). This term in turn will be very large compared with the second term in the first sum, provided that we are measuring power on scales considerably larger than δl. Thus, using $P_{1\mathrm{D}}(k) = \int dx\, \xi(x) \cos\,(kx) \to \sum \delta l \langle \Delta_i \Delta_j \rangle \cos\,[k(x_i - x_j)]$, we obtain

$$\langle f_k^2 \rangle \simeq \sum_i (\Phi_i^{(2)} - \Phi_i^2) + P_{1\mathrm{D}}(k) \sum_i \Psi_i^2/\delta l \,, \tag{C7}$$

and therefore we have an estimator for any extra one-dimensional power,

$$\hat{P}_{1\mathrm{D}}(k) = \left[\langle f_k^2 \rangle - \sum_i (\Phi_i^{(2)} - \Phi_i^2) \right] \Big/ \sum_i \Psi_i^2/\delta l \,. \tag{C8}$$

This should be compared with equation (3.30). There we found that the mean contribution to the one-dimensional power spectrum from small-scale clustering was

$$P_{1\mathrm{D}\text{-ss}}(k) = \frac{P_0}{2\pi} \frac{\int dz\, z^{2-n} \phi_g^2(z)}{\int dz\, z^4 \phi_g^2(z)} \frac{\int d\kappa\, \kappa^{1+n} \tilde{W}^2(\kappa)}{[\int d^2\theta\, W(\theta)]^2} \simeq \frac{283}{2\pi}\, h^{-1}\ \mathrm{Mpc} \tag{C9}$$

(with ϕ_g denoting the selection function for galaxies), whereas here we have

$$P_{1\mathrm{D}\text{-ss}}(k) = \sum_i (\Phi_i^{(2)} - \Phi_i^{(2)}) \Big/ \sum_i \Psi_i^2/\delta l \,. \tag{C10}$$

Interestingly, this estimator appears to be more robust than the traditional approach. In equation (3.30) we had to assume a correlation length and that the power spectrum was a power law, etc., with concomitant uncertainty. Here we effectively estimate the small-scale clustering properties internally.

We can also calculate the "coherence length" k_c which determines the band-averaged power fluctuations in a manner very similar to that of § 3.5. Neglecting the possible true long-wave power, we find (using the same approximations as before)

$$\langle f_k f_{k'} \rangle \simeq \sum_i (\Phi^{(2)} - \Phi^2) \exp\,[i(k - k')x_i] \,. \tag{C11}$$

Comparing this with equations (3.23), we see that the spatial fluctuation intensity is now $\mathscr{I}(r) = [\Phi^{(2)} - \Phi^2]/\delta l$ and, consequently, from equation (3.37) we have

$$k_c = \frac{2\pi \int dr\, \mathscr{I}^2(r)}{[\int dr\, \mathscr{I}(r)]^2} = \frac{2\pi \sum_i (\Phi^{(2)} - \Phi^2)^2/\delta l}{[\sum_i (\Phi^{(2)} - \Phi^2)]^2} \,. \tag{C12}$$

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1991ApJ...379..482K

We can now apply these results to our simple b = constant, clipped histogram weighting $[F(n) = 1$ if $n \geq 1; F(n) = 0$ if $n = 0]$. We have $\Phi_i^{(2)} + \Psi_i/b = \Phi_i$, this being just the (smoothed) fraction of occupied cells. We have estimated the sums appearing in equation (C10) numerically by filtering the clipped NGP + SGP histogram with a Gaussian window of scale length $100h^{-1}$ Mpc to obtain a smooth Φ_i, resulting in

$$P_{\text{1D-SS}}(k) = \sum_i (\Phi_i - \Phi_i^2) \Big/ b^2 \sum_i \Phi_i^2/\delta l \simeq 15.7 b^{-2} h^{-1} \text{ Mpc} . \tag{C13}$$

For $b = 1$ this is about a factor of 3 times smaller than expression (C9) above.

For the clipped histogram model the coherence length for the power spectrum is

$$k_c = \left[2\pi \sum_i (\Phi - \Phi^2)^2/\delta l\right] \Big/ \left[\sum_i (\Phi - \Phi^2)\right]^2 \simeq 2.7 \times 10^{-3} h \text{ Mpc}^{-1} . \tag{C14}$$

For the same survey (NGP + SGP) we found $k_c \simeq 8.3 \times 10^{-3} h$ Mpc^{-1}, so the coherence length here is about 3 times smaller, resulting in a further improvement in sensitivity of about $3^{1/2}$, or about a factor of $5b^2$ improvement in all.

What is the appropriate value for b? If we lay down grains in a Poissonian manner with mean density μ per cell, the fraction of occupied cells will be $1 - e^{-\mu}$, so $\mu = -\ln(1 - \Phi)$, and the fractional fluctuation in the occupied cell fraction due to a grain density perturbation $\delta\mu/\mu = \Delta$ will be $\delta\Phi/\Phi = b\Delta = \mu e^{-\mu}\Delta/(1 - e^{-\mu})$. For the BEKS survey we have $\Phi \simeq 75/200 = 0.375 \Rightarrow b \simeq 0.78$. The improvement in power sensitivity is therefore about a factor of 3.

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